The Price of Accountability: Democratic Institutions and the (In)Stability of Economic Data*

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Abstract

Political scientists often judge an economic statistic by its validity (how closely it tracks reality), neglecting its reliability (whether it reports the same value across releases). Yet reliability is political, not just technical. Revising a published statistic like GDP, while routine, generates the perception that public officials were wrong, inviting accusations of incompetence or manipulation. No government wants to pay this cost. Using 104 vintages of the World Development Indicators (1994–2021), I show that revisions are more frequent and larger for data collected by democracies, where press and academic freedom empower experts to scrutinize official statistics and demand corrections. This matters for research: the numbers reported for a democracy differ more across releases than those reported for an autocracy.

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1 Introduction

In August 2024, the United States Bureau of Labor Statistics (BLS) revised its preliminary employment data, finding that 818,000 fewer jobs had been created between March 2023 and March 2024 than initially reported. This scheduled annual benchmark revision came as no surprise to experts. Initial job estimates often rely on incomplete household and business surveys; as survey funding and response rates decline, the uncertainty around these estimates tends to increase. In relative terms, the revision was marginal: it meant that 159.2 million individuals, not 160 million, were employed in March 2024. However, in the wake of the revision, then-presidential candidate Donald Trump accused then-president Joe Biden of "fraudulently manipulating job statistics," an opinion echoed by several others in the Republican Party.[1]

One year later, in August 2025, the BLS again revised its preliminary employment data downward. That same day, President Trump fired the BLS commissioner, claiming on social media that the revisions had been "RIGGED in order to make the Republicans, and ME, look bad." Kevin Hassett, the director of the White House National Economic Council, complained that the BLS had been "revising numbers all over the place ... I don't think anybody really can trust that the numbers are right."[2]

The US is not unique in facing controversy over economic statistics. In 2009, Greece revised its planned deficit ratio from 3.7 to 12.5% of the Gross Domestic Product (GDP), triggering credit rating downgrades and a loss of market access that brought the country close to default (Aragão and Linsi, 2022). Between 2010 and 2014, Ghana, Kenya, Nigeria, and other African nations reported GDP increases of up to 60% after updating accounting methods and incorporating informal activities (Jerven and Ebo Duncan, 2012) — a shift associated with less generous lending terms, as countries with per capita incomes above a

[1] Alicia Wallace. "Trump Routinely Calls Economic Data 'Fake.' Here's Why That's Dangerous." *CNN.* 26 January 2025.

[2] Nick Niedzwiadek and Sam Sutton. "Trump Fires Statistics Chief After Soft Jobs Report." *Politico.* 1 August 2025.

certain threshold lose access to concessional lending (Kerner et al., 2017). In 2016, Ireland's reported GDP growth of 26.3% (later traced to Apple's onshoring of intellectual property) drew ridicule as "leprechaun economics" (Polyak, 2023). These cases may seem unrelated, but they share a common structure: in each one, a government revised its data to bring official statistics closer to economic reality, and paid a political price for doing so.

Measuring the economy is an iterative process: preliminary estimates are routinely revised to incorporate new data, improve methods, and correct mistakes (Carson et al., 2004). Macroeconomic indicators like GDP, trade, inflation, and unemployment come from many sources, and even the most advanced nations cannot get the "correct" numbers up front. Revisions are a statistical best practice and a natural, necessary component of the data production process. The International Monetary Fund (IMF), the World Bank, AFRISTAT, Eurostat, and other international organizations regularly assist countries seeking to improve their data collection and revision processes. The resulting changes are typically minor technical adjustments.

Data quality has two distinct dimensions. According to McMann et al. (2022), validity reflects how accurately data capture the underlying phenomenon being measured; reliability is the absence of unsystematic measurement error, the degree to which repeated applications of the same procedure yield consistent results.[3] Because each release re-estimates the same figures, disagreement across successive releases reflects precisely this inconsistency. Revisions improve validity by bringing official estimates closer to the truth. However, they undermine reliability: numbers that change from one release to the next cannot be trusted to mean the same thing. Governments that prioritize validity will update their data each time better information becomes available, even at the cost of reliability. Those prioritizing reliability may resist correction and sacrifice validity, keeping numbers unchanged to avoid any perception of instability. This trade-off is not just technical: it is political.

As the US case shows, even comparatively small, technical revisions can have substantial

[3]In McMann et al.'s framework, whether an indicator captures the same theoretical concept across reference years is a question of content validity, not reliability; my focus is on reliability.

political and economic consequences. Acknowledging past errors can undermine the government's reputation, imply a low commitment to transparency, trigger market instability, and erode public trust. The general public often misunderstands the purpose of revisions, and opposition parties capitalize on this misunderstanding. As Ghana, Kenya, and Nigeria show, even large *upward* revisions can be costly, weakening the government's bargaining position in international financial negotiations. Whether large or small, upward or downward, revisions are both a technicality and a political liability. Under what conditions do the technical pressures to revise outweigh the political costs?

Producing official statistics is a two-stage process. Governments first collect and release preliminary estimates, then decide whether to revise the preliminary estimates (a choice shaped by the quality of the original data). I argue that the context of data *collection* is more important than the context of data *revision*. On average, countries with higher state capacity have better-trained statisticians, regular household surveys, and comprehensive administrative data systems that let them collect and report higher-quality data on time, setting the stage for future revisions. While state capacity provides the technical ability to detect and incorporate new information, governments must also be willing to revise. No government welcomes the political costs that revisions impose, but I argue that revisions are more likely when accountability pressures make the cost of inaction higher than the cost of correction.

These accountability pressures operate primarily through domestic institutions. Democracies tend to have independent statistical offices that publish their data sources and methodology. This, coupled with academic freedom and free speech, empowers experts to scrutinize official records, identify discrepancies, and demand corrections. Through media coverage, public debate, and academic research, experts compel democratic governments to acknowledge errors and correct official estimates. Democracies do not necessarily *want* to revise their data. Still, journalists, academics, and other experts *pressure* democracies to make technically necessary but politically costly revisions that would otherwise be overlooked.

Opposition parties may try to politicize statistical controversies, but it is the credibility of independent experts — not just political competition — that drives governments to undertake revisions.

Most of the existing research focuses on data validity (e.g. Martínez, 2022; Magee and Doces, 2015), asking under what circumstances data capture underlying theoretical concepts with precision and minimal error. My contribution is to examine the second, less explored dimension: reliability.[4] Instead of treating validity and reliability as independent properties, I show that they are in tension: the conditions that produce more valid data are also the conditions that produce less reliable data.

First, I outline the threats to validity that create demand for revisions — *why* countries revise their data. Second, I develop a theory of reliability that explains the timing of revisions — *when* countries revise their data. I find support for my theory in an analysis of GDP data published by the World Development Indicators (WDI) between 1994 and 2021: the democracy finding is robust across three measures of regime type (Polyarchy, Polity 2, and Freedom House) and holds for both the likelihood and magnitude of revisions. I conclude by discussing research and policy implications.

My findings underscore the trade-off between validity and reliability, the credibility risks associated with data revisions, and the understudied consequences of revisions to the same data source. While inconsistencies *across data sources* have been widely documented (Linsi et al., 2023; Weikmans and Roberts, 2019; Ram and Ural, 2014; Kerner, 2014; Michaelowa and Michaelowa, 2011), I show that inconsistencies *across releases of the same source* may be even more consequential (Goes, 2023; Johnson et al., 2013). Because democracies revise more, the numbers reported for a democracy differ more across releases than those reported for an autocracy. Data reliability is not a neutral technical property but a political outcome shaped by the same institutional forces that determine whether governments are willing to be held accountable at all.

[4]See Fariss et al. (2022) for one of the few attempts to quantify the reliability of GDP, GDP per capita, and population measurements.

2 Why Revise?

There are three types of data revisions: information-based, structural, and error correction (Croushore and Stark, 2003). The first two are routine in macroeconomic data production, whereas the third signals that something has gone wrong. I briefly describe all three below, with more detailed information in Appendix C.

Most revisions are information-based: they are related to the "discovery" of new data. As the 2024 BLS employment data revision illustrates, preliminary estimates tend to be published before all source data are available. As household surveys, tax records, and other administrative data are compiled, international experts advise countries to revise estimates following a regular, publicized schedule that clearly distinguishes between preliminary and updated data.[5] The reasoning is that timely (if incomplete) macroeconomic data improve governance by enabling more informed policymaking (Islam, 2006). For example, the IMF-produced Report on the Observance of Standards and Codes (ROSC) Data Module[6] praises countries that revise regularly and reprimands those that do not: "Studies and analyses of revisions should be conducted routinely and used to inform statistical processes and data users" (Sweden, 2001). "Even preliminary data, with the understanding that these are subject to revisions, would be useful" (Uruguay, 2001). These revisions are typically small, expected, and most common within one year of the initial data release (Croushore and Stark, 2003).

Revisions can also be structural, following deliberate changes to definitions, base years, or aggregation methods. The System of National Accounts (SNA), a global standardization framework, is updated periodically; the 1993 SNA introduced the concept of imputed rent,

[5] As of 2024, 95% of IMF members subscribe either to the Special Data Dissemination Standard (SDDS) or the enhanced General Data Dissemination System (e-GDDS). SDDS subscribers must disseminate preliminary national accounts data with a lag of no more than one quarter after the end of the reference period. Subscribers to the less demanding e-GDDS have no hard cutoff. These preliminary estimates should not be confused with flash or advance estimates, which many industrialized nations release within 30 days of the end of each quarter, based on incomplete data.

[6] Most ROSC Data Modules were conducted between 1999 and 2010, and 98% of them have been published. Published reports are available at https://www.imf.org/en/Publications/rosc?sortBy=Topic&sortVal=Data%20Dissemination

which significantly altered the definition and calculation of GDP (Olinto Ramos et al., 2008). In 2014, EU countries implemented new Eurostat accounting standards based on the 2008 SNA, revising their GDP definitions to include illegal activities such as drug trafficking and prostitution. As a result, the Italian and British economies increased by 4% each (Coyle, 2014, 110). The longer the time since the original data release, the more structural the revisions (Croushore and Stark, 2003). Structural revisions reflect an evolving consensus about what the indicator should capture.

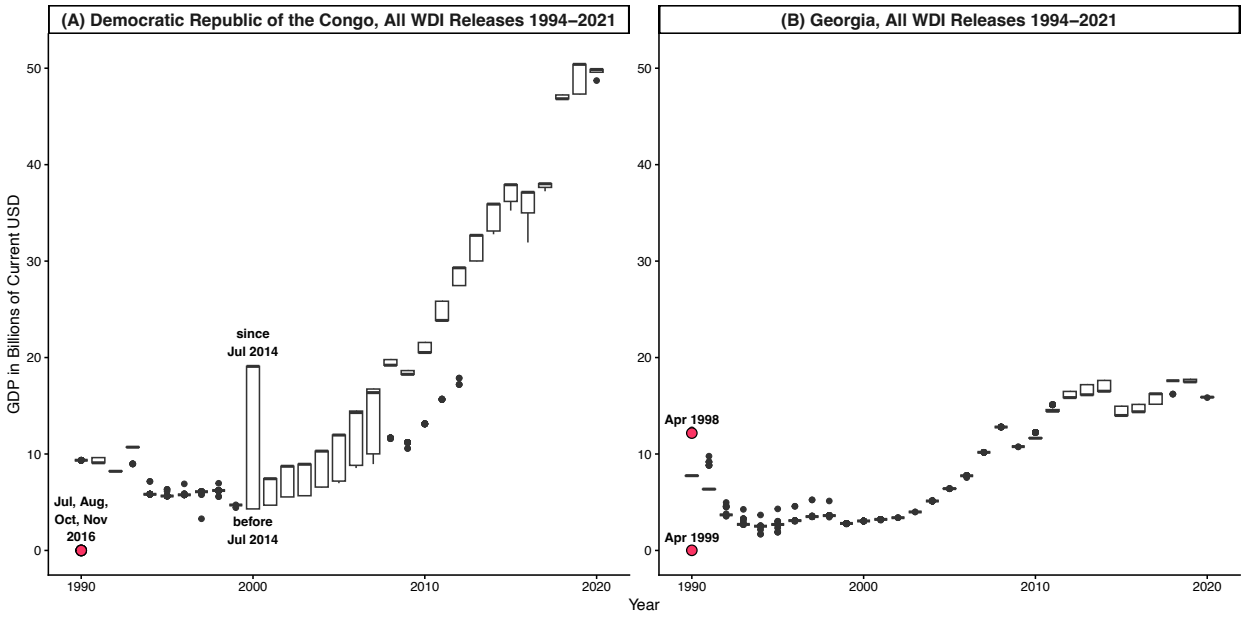
Finally, revisions correct errors in preliminary data. Three factors commonly make data "incorrect" enough to require revision: low statistical capacity, strong political incentives, and inadequate data management. Regarding statistical capacity, national statistical offices (NSOs) are often underfunded, understaffed, use outdated methods, experience frequent turnover, and rely on outdated information about businesses and households. Many of their employees lack basic statistical competence or familiarity with the SNA (Mejía Guerra et al., 2023). Together, these factors limit their ability to collect, standardize, and disseminate high-quality data.

Political incentives also explain "incorrect" data. Autocracies are less likely to report policy-relevant data (Hollyer et al., 2011) and may overstate growth rates or underreport COVID-19 deaths (Martínez, 2022; Magee and Doces, 2015; Wallace, 2014; Adiguzel et al., 2020). Political appointees at NSOs might face dismissal if their statistics displease leaders. After the BLS reported unfavorable employment data in August 2025, President Trump dismissed its commissioner and appointed a replacement who proposed ending monthly jobs reports.[7] Fiscal transfers, aid dependence, climate-conscious electorates, and EU rules create incentives to inflate population figures, understate economic growth, overstate climate aid, and violate deficit limits, respectively (Devarajan, 2013; Kerner et al., 2017; Michaelowa and Michaelowa, 2011; Alt et al., 2014). Conversely, political competition and frequent turnover increase uncertainty about the future, motivating incumbents to pass Freedom of Information

[7]Natalie Sherman. "Trump's Pick to Lead Economic Data Agency Floats Ending Monthly Jobs Report." *BBC*. 12 August 2025.

(FOI) laws that enhance transparency (Berliner, 2014).

Figure 1: Current GDP of the Democratic Republic of the Congo and Georgia, 1990–2020



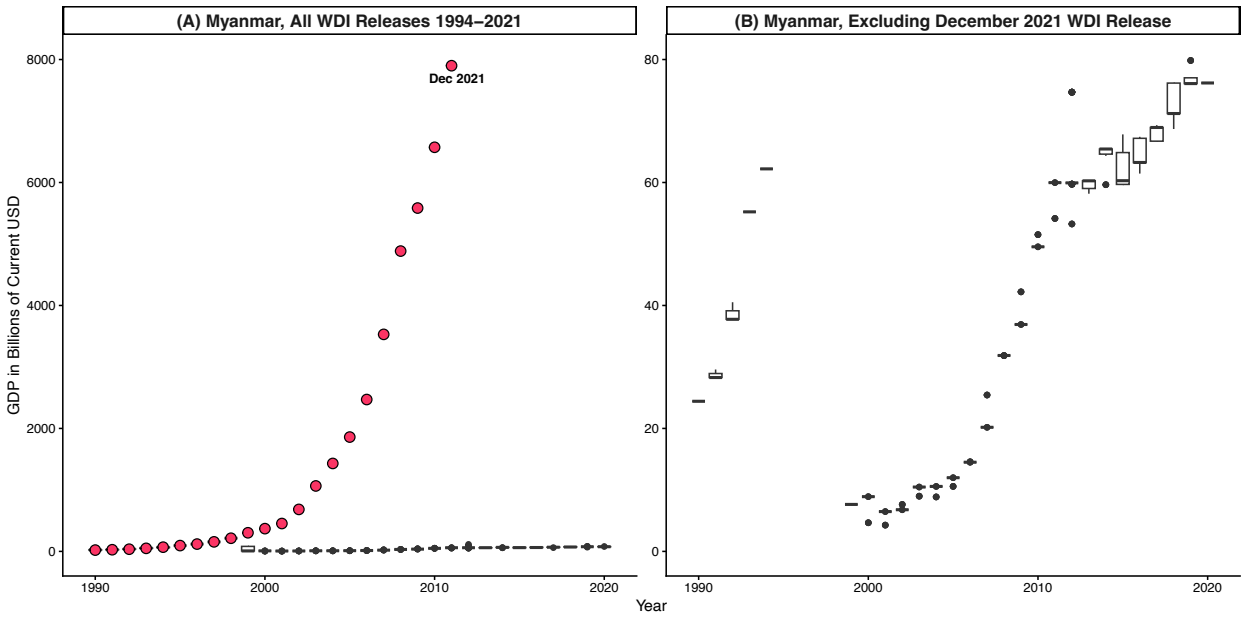
These boxplots present the distribution of current GDP estimates from 1990 to 2020 for (A) the Democratic Republic of the Congo and (B) Georgia, using data drawn from the 104 WDI releases from April 1994 to December 2021. Four WDI releases (in pink) reported a GDP of zero for the Democratic Republic of the Congo in 1990. All 32 releases before July 2014 reported a GDP of 4.3 billion for 2000, a figure revised to 19.1 billion in July 2014. Georgia's GDP in 1990 was reported as 12.1707 *million* in some WDI vintages and 12.1707 *billion* in others (in pink). Section 4.1 describes the data in detail.

Idiosyncratic errors present a final threat. As the pink dots in Figure 1 show, four WDI vintages report the Democratic Republic of the Congo's 1990 GDP as *zero*; two others report it as missing. Georgia's 1990 GDP — reported to be around 12.1707 *billion* current US dollars until April 1998 — "lost" three digits in the April 1999 and April 2000 vintages, shrinking to 12.1707 *million*.[8] In the December 2021 update, Myanmar's 2011 GDP was over 100 times higher than in any other vintage: 7.899 *trillion*, versus 54 to 59 *billion* (Figure 2). Other minor errors include swapped country names, mislabeled columns, and entire indicators mistakenly blanked (see World Bank 2023 for errata).

Overall, the underlying assumption is that revisions *increase* the validity of preliminary,

[8]Though Georgia gained formal independence from the Soviet Union in December 1991, its WDI coverage begins in 1990.

Figure 2: Current GDP of Myanmar, 1990–2020



These boxplots present the distribution of current GDP estimates from 1990 to 2020 for Myanmar, using data drawn from the 104 WDI releases from April 1994 to December 2021. The December 2021 WDI release (in pink) is included in (A), but not in (B). As the different y-axes show, the December 2021 release was an outlier, reporting exceptionally high values for the entire time series. Section 4.1 describes the data in detail.

imperfect data. This assumption could be wrong: revisions could *decrease* validity. However, governments rarely seem to falsify data ex post (Guriev and Treisman, 2019). Political leaders might withhold statistics (Venezuela since 2015), postpone the initial release (Zimbabwe in 2019), release doctored numbers from the outset (Argentina in 2007–2008), or erase already-published statistics from public records (the US in 2025), but do not tend to replace "correct" information with falsified information.

In sum, revision is a normal part of data production. Yet governments often resist even routine updates. This resistance is not arbitrary. Revisions carry real political costs: they signal that previous estimates were wrong, invite accusations of incompetence or manipulation, and give opposition parties a ready-made narrative of governmental failure. Although information-based and structural revisions are analytically distinct from error correction, the public rarely knows the difference, instead viewing every revision as an admission of error or

bad faith. A government that never revises its data avoids these costs entirely. Its numbers may be less valid, but they are consistent — and consistency has its own political value. Even the international bodies that encourage revision recognize its reputational risks, warning that large corrections can signal a lack of "independence, integrity and accountability of the national statistical authorities" (European Commission, 2010, 3).

The puzzle is not why governments produce imperfect data. Given the challenges of measuring a modern economy, imperfect data are inevitable. The puzzle is why some governments correct their imperfect data while others do not. If revisions are both statistically necessary and politically costly, something must overcome governments' baseline preference for inertia. The next section develops a theory of when and why that happens, arguing that the answer lies not in the moment of revision itself, but in the institutional conditions under which the original data were collected.

3 How Accountability Pressures Shape Data Revisions

What overcomes governments' baseline preference for inertia? I argue that the answer lies in accountability pressures that raise the political cost of *not* revising above the political cost of revising. Still, accountability pressures interact with a two-stage data production process in which the conditions of the first stage (when data are initially collected and disseminated, in reference year y) shape the feasibility and likelihood of corrections in the second stage (when previously published data may be revised, at a later vintage year v). This sequencing matters: strategies initially adopted at y are hard to reverse at v, because they determine what can be corrected later. A government that collects opaque, poorly documented data forecloses meaningful revision later, however strong the pressures it faces. A government that collects transparent, well-documented data makes revisions both technically feasible and externally demanded.

State capacity sets the foundation. The capacity to produce reliable statistics rests on

what Lee and Zhang (2017) call legibility: the state's ability to render social and economic activity into standardized, administrable information. Countries with higher state capacity can train and retain skilled personnel, conduct regular surveys, digitize administrative records, audit past estimates, and facilitate cooperation between agencies (for example, between the NSO, tax agency, and finance ministry). All this enhances the quality of initial data, making subsequent revisions easier to implement. Capacity determines whether revision is technically feasible, but not whether governments are willing to bear its political costs. Several high-income nations that could afford to release and revise their data choose not to (Williams, 2009). A report by the African Development Bank (2013, 5) underscores this point: "rich countries are not guaranteed to have good statistics; neither are poor countries condemned to have bad data ... Achieving high-quality data entails a political choice and a firm commitment to invest in statistics." Capacity does not determine whether that political choice is made; accountability pressures do.

Democracies and constrained executives release more data (Hollyer et al., 2011; Williams, 2009) and of higher validity (Magee and Doces, 2015; Martínez, 2022). Leaders facing stiff political competition and regular turnover are more likely to institutionalize transparency through FOI laws, securing future access to government information (Berliner, 2014). According to Brambor et al. (2020), democracies are better at collecting and processing data not because of political competition, but because of expanded suffrage: states must collect fine-grained information to enable broad popular participation. Democracy is associated not only with more and better data, but also — I argue — with a greater likelihood of data revisions. Beyond the *ability* to revise, countries must be *compelled* to do so. Democracy is the primary institutional mechanism for raising the political cost of inertia and overcoming governments' reluctance to revise. Democracies enable the work of experts who serve as accountability agents, pressuring for regular revisions no matter how high the political cost.

Democracies are more willing to publicly release raw data, codebooks, and data collection protocols; to retain experienced data managers rather than replace them with political

appointees; and to establish external advisory committees that provide technical guidance and independent oversight. This makes their data easier to correct at time v. This does not happen due to an intrinsic commitment to transparency. Given the high political cost of data corrections, no government — democratic or authoritarian — would like to admit its mistakes. Still, democracies have little choice: their institutions create incentives to release transparent data ex ante and ongoing pressure to revise these data ex post.

In terms of the mechanism, I expect revisions to be driven not by political competition but by technocratic expertise. Through media coverage, public debate, and academic research, experts continuously pressure democratic governments to acknowledge errors and correct official estimates. One possibility is that democracies face such intense transparency pressures that they rush to release incomplete data, creating a greater need for subsequent revisions. Another possibility is that they are more cautious and initially publish understated estimates, unlike dictators, who — according to Martínez (2022) — tend to exaggerate their numbers. Either way, democratic leaders have limited control over revisions, which will be as frequent and as large as experts deem necessary.

For the same reason, democracy affects not only the frequency but also the magnitude of revisions. Since democratic institutions empower experts to demand corrections, governments cannot limit revisions to small, face-saving adjustments. When the data requires large corrections, experts will push for them.

In contrast, autocracies are less likely to disclose their data sources and methodology. Without checks and balances, the central government might tamper with the NSO's work, preventing the collection of important evidence. Autocracies like Azerbaijan, Belarus, Oman, or Tajikistan might recognize the instrumental value of data transparency and request ROSC Data Modules, yet the reputational rewards of international standards rarely outweigh the domestic cost of admitting error. In these countries, data releases are strategic and selective — for example, to attract development aid and foreign investment, gain or maintain access to capital markets, monitor internal challenges, and allocate resources to secure elite support

(Hollyer et al., 2018). "Informational autocrats" gain legitimacy by disseminating selective information about economic successes while concealing information about economic failures (Guriev and Treisman, 2019). They prioritize control over timeliness, delaying or avoiding publication of data deemed politically unsafe. Information collected by autocrats is a black box: the opaque data-generation process and centralized information control leave little room for subsequent public scrutiny. When freedom of speech is limited, journalists and academics may hesitate to question official statistics or push for corrections. In particular, autocracies with low state capacity face neither the technical conditions that make revision feasible nor the accountability pressures that make it necessary.

Hypothesis 1: The more democratic a country is at reference year y, the more likely its data for y are to be revised.

Hypothesis 2: The more democratic a country is at reference year y, the larger the revision, when one occurs.

My theory is agnostic about the underlying validity of the revised data. Some countries might have rushed to release low-quality preliminary information at time y that must be corrected ex post at time v. Others might only need to fine-tune their data, having released high-quality information from the start. Whatever the validity, I argue that contextual factors at the time of data collection, y, best explain revisions. Conditional on state capacity, domestic accountability pressures incentivize releasing transparent information from the outset, enabling subsequent revisions.

4 Operationalizing Data Revisions

4.1 Outcome: Current GDP

As the most widely used source of macroeconomic data in political science (Goes, 2023), the WDI Database Archives provide 104 electronic releases from 1994 to 2021, each capturing the same underlying data at a different point in time.[9] The differences between snapshots are the revisions I seek to explain.

GDP, the value of all final goods and services produced in a country during a specific period, is "the superstar of indicators" (Hoekstra, 2019, 6) and the most ubiquitous measure of national wealth. In 2020, 205 out of 206 economies surveyed by the IMF compiled and published annual GDP statistics (Baer et al., 2022); Eritrea was the lone exception. By comparison, only 109 compiled institutional sector accounts, such as deficit, debt, trade, and foreign direct investment (FDI). Therefore, I focus on revisions to GDP data — specifically, the indicator *GDP in current US dollars* (ID `NY.GDP.MKTP.CD`), the annual "sum of gross value added by all resident producers in the economy."[10] The production approach is the most widely used and disseminated method for GDP estimation (Baer et al., 2022, 12). Current GDP enables comparisons across releases, though not across countries or over time, because it does not adjust for purchasing power parity (PPP) or inflation.[11]

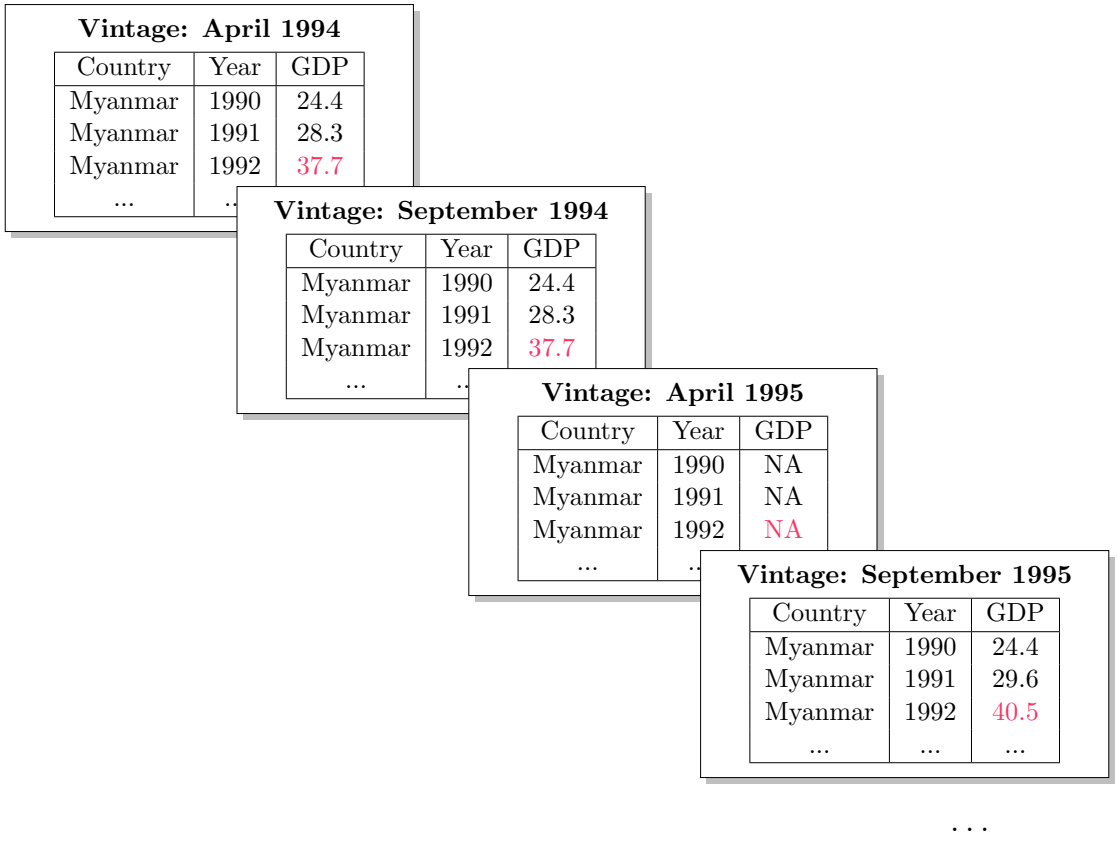
There are two stages of data production: the reference year, when country c collects and disseminates its data ($y = 1990, \ldots, 2020$), and the WDI vintage, when the same country may

[9] Though all releases since 1989 are available, the indicator of interest is missing from all releases before 1994, and no release is available for 1996.

[10] GDP in US dollars is converted from local currency units (LCU) at market exchange rates, raising the concern that revisions could reflect exchange rate movements rather than re-estimated outputs. However, a replication in local currency is not feasible: *GDP (current LCU)* (ID `NY.GDP.MKTP.CN`) is distorted by currency changes that cannot be separated from genuine revisions, while *GDP: linked series (current LCU)* (ID `NY.GDP.MKTP.CN.AD`) smooths over the structural breaks I study and is missing for 70% of the observations. In contrast, the series I use absorbs these changes while preserving genuine revisions. In any case, the Bank is unlikely to revise exchange rates differently for democracies and autocracies.

[11] *GDP, PPP (current international $)* (ID `NY.GDP.MKTP.PP.CD`) allows for comparisons across countries, but not across releases, as the PPP conversion factor is periodically revised. *GDP in constant US dollars* (ID `NY.GDP.MKTP.KD`), calculated using the GDP deflator to account for inflation, allows for comparisons over time, but not across releases.

Figure 3: Illustration of the Data Structure, Using the Case of Myanmar



This figure uses the case of Myanmar to illustrate the three-dimensional structure of the data: the unit of analysis is country c, year y, and vintage v. In this notation, $y = 1990, \ldots, 2020$ refers to the first stage, when the data are initially collected and disseminated, whereas $v =$ April 1994, September 1994, ..., December 2021 refers to the second stage (i.e., each subsequent WDI release), when previously published data may be revised.

revise its data ($v =$ April 1994, September 1994, ..., December 2021). Figure 3 illustrates this structure using Myanmar's GDP in billions of current US dollars as an example. In this figure, each vintage corresponds to a separate spreadsheet containing all available reference years.[12] The goal of my analysis is to explain the differences between spreadsheets.

For country c and reference year y, I define a revision as any change in GDP between two consecutive WDI vintages, $v-1$ and v. $Revision_{c,y,v}$ can reflect an addition, deletion, or

[12]To ensure that each reference year has had time to be compiled and published, the analysis includes only vintages released at least 90 days after the end of the reference period. For example, for the reference year 2020, this corresponds to all WDI releases starting in April 2021. I use 90 days as a cutoff because 39.8% of all economies surveyed by the IMF in 2020 reported annual GDP data within 90 days of the reference period; 54.4% did so within 91 to 365 days (Baer et al., 2022, 17).

change of any magnitude, in any direction. As Figure 3 illustrates, Myanmar's 1992 GDP was reported as 37.7 billion in April 1994, 37.7 billion in September 1994 (*Revision = 0*), missing in April 1995 (*Revision = 1*), and 40.5 billion in September 1995 (*Revision = 1*). Revisions occur in 16.95% of all observations.

Among revisions where neither value is missing, I also examine relative size. These revisions are overwhelmingly small: the median revision changes GDP by 0.00015%, and three-quarters of revisions are under 0.4%. The mean, 4.4%, is much larger, inflated by a long tail of rare revisions ranging from a 100% reduction (the Democratic Republic of the Congo, Figure 1) to a 17,488% increase (Myanmar, Figure 2). Because the distribution is so skewed, I use the absolute value of percentage changes, logged to reduce the influence of these extremes: $\textit{Absolute \% Change (Log)}_{c,y,v}$. Revisions are roughly balanced in direction, with 55.6% revising GDP upward and 44.4% downward. A revision may also involve a value appearing or disappearing — a country-year missing in one vintage and reported in the next, or vice versa. Since a percentage change cannot be computed in these cases, they count as revisions in the likelihood analysis but are excluded from the magnitude analysis.

4.2 Independent Variables

My theory focuses primarily on regime type as the main predictor of data revisions, with state capacity serving as a background condition. I measure regime type using three indices. My primary measure, V-Dem's *Polyarchy* index (Coppedge et al., 2023), captures the quality of electoral democracy (including extensive suffrage, fair elections, freedom of expression, and access to information) on an ordinal scale from 0 to 1. As robustness checks, *Polity 2* (Marshall and Gurr, 2020) aggregates institutional constraints into executive authority, ranging from –10 (hereditary monarchy) to +10 (consolidated democracy); *Freedom House* combines political rights and civil liberties into a 7-point score, from 1 (most free) to 7 (least free), a scale I flip so it is consistent with the other two measures. Across all three indices, higher values indicate more democratic regimes (and thus more domestic accountability

pressures). Still, the three indices are not interchangeable. Polyarchy explicitly incorporates freedom of expression and access to information, key components of my theory. Polity 2 weights institutional constraints on the executive more heavily. Freedom House captures expression-related rights, but as a small share of a composite that also covers political rights, associational rights, personal autonomy, and rule of law. Correspondingly, I expect the association to be strongest for Polyarchy.

Why are democracies more subject to public scrutiny? The key mechanisms I propose are *Media Freedom* and *Freedom of Academic Exchange*. The former captures how much governments censor political information in the media; the latter captures how free scholars are to exchange and communicate research findings, including through media engagement and public lectures. Both are V-Dem indicators (Coppedge et al., 2023), from 0 to 4, with higher values indicating rare censorship and full academic freedom, respectively.[13] Without censorship and intimidation, experts can scrutinize official statistics and demand transparent data practices. I also assess alternative mechanisms from Berliner (2014): *Opposition Strength* measures the vote share of the largest opposition party in the most recent legislative election, whereas *Turnover Frequency* indicates the number of changes in executive party control over the preceding five years (both from the Database of Political Institutions, Cruz et al. 2021). Following Brambor et al. (2020), who show that suffrage expansion drives investments in information capacity, I include the population share with *Suffrage* as an alternative pathway through which democracy might affect revision behavior through broad popular participation. Finally, I include the *Political Corruption Index* from V-Dem as an alternative mechanism: more corrupt environments may suppress the magnitude of revisions, as officials have incentives to minimize the visibility of corrections rather than acknowledge past errors.

My main proxy for state capacity is *Public Administration*, a V-Dem measure ranging from 0 (public officials do not respect the law) to 4 (public officials generally fully respect the

[13]The official variable names in the V-Dem codebook are *Government Censorship Effort — Media* and *Freedom of Academic Exchange and Dissemination*.

law).[14] This measure is imperfect: it primarily captures bureaucratic impartiality and rule-following, rather than the informational capacity — the ability to render economic activity legible through standardized records and surveys (Lee and Zhang, 2017) — that most directly underpins statistical production. Still, it has broader temporal and spatial coverage than alternative capacity measures, such as the Hanson and Sigman (2021) state capacity index, the Brambor et al. (2020) information capacity index, and the International Country Risk Guide measure of bureaucratic quality.[15]

Models also include a dichotomous indicator of *IMF Program* participation (Kentikelenis and Stubbs, 2023) as a control for foreign accountability pressure, along with indicators for a *Financial Crisis* (Nguyen et al., 2022), *Natural Disaster* (Centre for Research on the Epidemiology of Disasters, 2020), and *Armed Conflict* (Gleditsch et al., 2002), all of which may independently prompt data revisions.

Three final variables capture technical or methodological factors. To capture structural revisions from methodological updates, *SNA Change* equals one in years when countries adopted or updated the SNA in use, using data from the UN National Accounts Statistics (complemented by WDI Metadata and IMF International Financial Statistics). To capture idiosyncratic mistakes, *Data Management Error* takes the value of one for reporting mistakes documented in the World Bank's data updates and errata (World Bank, 2023), like those in Figures 1 or 2 (see Appendix C). Lastly, *Log Difference* measures the gap between v and y, logged to capture the nonlinear decay in revision frequency over time. Information-based revisions are most common shortly after the initial data release. As time passes, revisions become predominantly structural (Croushore and Stark, 2003).

Table 1 summarizes all independent variables and their underlying concepts. All substantive predictors are lagged by one year (to avoid simultaneity bias) and assigned to the first stage (y), consistent with my core argument that the conditions under which data

[14] The official variable name in the V-Dem codebook is *Rigorous and Impartial Public Administration*.

[15] As Appendix D shows, the democracy result is robust to replacing *Public Administration* with these alternative measures.

Table 1: Summary of Independent Variables

Variable	Underlying Concept	Period
Polyarchy	Domestic Accountability	$y-1$
Polity 2	Domestic Accountability	$y-1$
Freedom House	Domestic Accountability	$y-1$
Domestic Accountability Mechanisms:		
– *Media Freedom*	Mechanism: Media Freedom	$y-1$
– *Freedom of Academic Exchange*	Mechanism: Academic Freedom	$y-1$
– *Political Corruption Index*	Alternative Mechanism	$y-1$
– *Suffrage*	Alternative Mechanism	$y-1$
– *Opposition Strength*	Alternative Mechanism	$y-1$
– *Turnover Frequency*	Alternative Mechanism	$y-1$
Public Administration	State Capacity	$y-1$
IMF Program	Foreign Accountability	$y-1$
Financial Crisis	Exceptional Events	$y-1$
Natural Disaster	Exceptional Events	$y-1$
Armed Conflict	Exceptional Events	$y-1$
SNA Change	Methodological Changes	v
Data Management Error	Idiosyncratic Errors	v
Log Difference	Time Since Initial Data Dissemination	$v-y$

are initially collected matter more than the conditions at the time of potential revision.[16] Vintage-specific variables (SNA changes, data management errors, and time between vintage and year) are assigned to the second stage (v), since they capture technical features of the revision process itself, rather than the institutional environment of data collection.[17]

4.3 Empirical Strategy

I estimate two sets of models by maximum likelihood. First, I use logistic regressions to capture $Revision_{c,y,v}$, the probability that a country-year observation (say, Myanmar-1992) is revised between two consecutive WDI vintages:

$$\begin{aligned}\Pr(\text{Revision}_{c,y,v}=1) = \text{logit}^{-1}(\ &\alpha + \beta\,\text{Democracy}_{c,y-1} + \mathbf{X}_{c,y-1}\boldsymbol{\gamma} \\ &+ \mathbf{Z}_{c,v}\boldsymbol{\delta} + \lambda\,\text{Log Difference}_{c,v-y} + u_c + u_y + u_v)\end{aligned} \tag{1}$$

[16]The main analysis does not include the same predictors at both stages, as this would introduce severe multicollinearity (for example, *Polyarchy* values at y and v are correlated at $r = 0.88$, $p < 0.001$). Section 5.3 examines the two stages jointly to test the timing of the association; because of this collinearity, those models are interpreted through marginal effects rather than individual coefficients.

[17]I do not include economic indicators like foreign aid, natural resource rents, or trade dependence because they are revised in parallel with GDP.

Among revisions where neither value is missing, I also examine relative size., I use linear regressions to estimate its magnitude, *Absolute % Change (Log)*$_{c,y,v}$:

$$\begin{aligned}\text{Absolute \% Change (Log)}_{c,y,v} = \alpha + \beta\,\text{Democracy}_{c,y-1} + \mathbf{X}_{c,y-1}\boldsymbol{\gamma} \\ + \mathbf{Z}_{c,v}\boldsymbol{\delta} + \lambda\,\text{Log Difference}_{c,v-y} + u_c + u_y + u_v + \varepsilon_{c,y,v}\end{aligned} \tag{2}$$

Common to both equations, *Democracy*$_{c,y-1}$ is the regime measure, whereas $\mathbf{X}_{c,y-1}$ represents the remaining collection-stage covariates, measured at y and lagged one year; $\mathbf{Z}_{c,v}$ represents the vintage-stage covariates, measured contemporaneously at the vintage v; *Log Difference*$_{c,v-y}$ is the logged number of years between the reference year and the vintage; and u_c, u_y, u_v are random intercepts for country, reference year, and vintage, respectively. Equation 2 adds an idiosyncratic error $\varepsilon_{c,y,v}$. This two-step approach separately models whether a revision occurs and, if so, its size.[18]

5 Determinants of Data Revisions

5.1 Main Results

Table 2 presents the results of six models. Models 1 to 3 are logistic regressions with the dichotomous outcome *Revision*. Models 4 to 6 are three linear regressions with the continuous outcome *Absolute % Change (Log)*, restricted to instances when *Revision* equals one (hence the smaller sample). Each pair of models uses a different measure of regime type: *Polyarchy* (Models 1 and 4), *Polity 2* (Models 2 and 5), and *Freedom House* (Models 3 and 6).

Consistent with Hypothesis 1, higher first-stage democracy is associated with a higher likelihood of second-stage revisions. This association is positive and significant ($p < 0.01$) for all three democracy measures: *Polyarchy*, *Polity 2*, and *Freedom House*. The association is incremental: holding other covariates at their means, moving from 0.25 to 0.75 on the

[18] I model the two steps separately because Tobit or Heckman selection models assume independent observations and cannot incorporate random intercepts.

Table 2: Determinants of the Likelihood and Magnitude of Data Revisions

	Dependent Variable:					
	Revision	Revision	Revision	Abs. % Change	Abs. % Change	Abs. % Change
	(1)	(2)	(3)	(4)	(5)	(6)
Polyarchy, $y-1$	0.56^{***}			1.31^{***}		
	(0.09)			(0.34)		
Polity 2, $y-1$		0.02^{***}			0.02^{*}	
		(0.00)			(0.01)	
Freedom House, $y-1$			0.04^{***}			0.09^{*}
			(0.01)			(0.05)
Public Administration, $y-1$	0.04^{*}	0.05^{***}	0.07^{***}	0.12	0.19^{***}	0.17^{**}
	(0.02)	(0.02)	(0.02)	(0.08)	(0.07)	(0.07)
IMF Program, $y-1$	0.02	0.02	0.04^{**}	-0.14^{*}	-0.10	-0.10
	(0.02)	(0.02)	(0.02)	(0.08)	(0.08)	(0.08)
Financial Crisis, $y-1$	-0.00	-0.00	-0.00	-0.16^{**}	-0.18^{**}	-0.15^{*}
	(0.02)	(0.02)	(0.02)	(0.08)	(0.08)	(0.08)
Natural Disaster, $y-1$	0.05^{***}	0.06^{***}	0.06^{***}	-0.11	-0.13^{*}	-0.09
	(0.02)	(0.02)	(0.02)	(0.07)	(0.07)	(0.07)
Armed Conflict, $y-1$	0.05	0.04	0.05	-0.24^{*}	-0.22	-0.24^{*}
	(0.03)	(0.03)	(0.03)	(0.13)	(0.13)	(0.13)
SNA Change, v	0.36^{***}	0.35^{***}	0.36^{***}	1.26^{***}	1.36^{***}	1.25^{***}
	(0.03)	(0.03)	(0.03)	(0.11)	(0.11)	(0.11)
Data Management Error, v	6.28^{***}	6.36^{***}	6.28^{***}	11.52^{***}	11.87^{***}	11.29^{***}
	(1.16)	(1.16)	(1.16)	(1.49)	(1.52)	(1.53)
Log Difference, $v-y$	-1.02^{***}	-1.03^{***}	-1.00^{***}	-4.66^{***}	-4.71^{***}	-4.65^{***}
	(0.03)	(0.03)	(0.03)	(0.10)	(0.11)	(0.10)
Intercept	-2.48^{***}	-2.25^{***}	-2.18^{***}	-0.26	0.44	0.55
	(0.56)	(0.56)	(0.56)	(0.92)	(0.90)	(0.93)
AIC	148376.39	140484.83	147545.67	420299.73	398692.97	417780.71
Log Likelihood	-74175.19	-70229.41	-73759.84	-210135.86	-199332.48	-208876.35
Observations	375247	356493	373324	63050	59815	62669
Number of Countries	169	163	169	169	163	169
Number of Reference Years	31	30	31	31	30	31
Number of Vintages	103	103	103	78	78	78
Variance: Country (Intercept)	0.23	0.24	0.25	3.11	3.42	3.33
Variance: Reference Year (Intercept)	0.04	0.05	0.03	12.62	11.08	12.34
Variance: Vintage (Intercept)	27.89	27.51	27.93	26.17	26.35	26.02

Notes: This table presents three logistic regressions (Models 1–3) and three linear regressions (Models 4–6) with random intercepts for country, year, and vintage. y is the reference year; v is the vintage. Standard errors in parentheses. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

Polyarchy scale (roughly the gap between Russia and South Africa in 2020) is associated with about a one-third increase in the likelihood of revision. Higher democracy levels are also associated with larger revision magnitudes, though this result is weaker for *Polity 2* ($p = 0.061$) and *Freedom House* ($p = 0.051$). The magnitude finding is strongest for *Polyarchy*

($p < 0.01$): the same shift from 0.25 to 0.75 is associated with revisions roughly twice as large. The results support Hypothesis 2, though the strength varies across measures. As Section 5.3 shows, however, the magnitude association is concentrated among country-years that are also democratic at the vintage, whereas the likelihood association holds regardless of the vintage regime.

Among the controls, *Public Administration* is positively associated with the likelihood of revision, consistent with the expectation that higher bureaucratic capacity facilitates the detection and correction of errors. IMF programs, financial crises, natural disasters, and armed conflicts are generally associated with an increase in the likelihood of revision and a decrease in the magnitude of revision, reflecting disruptions to regular data collection during exceptional events. At time v, SNA changes and data management errors are consistently associated with a higher likelihood of revisions and larger revision magnitudes, as expected. The log difference between vintage and year is negative and highly significant in all six models, confirming that revisions are most common shortly after the initial data release and become progressively less frequent as more time elapses (Croushore and Stark, 2003). Finally, the vintage random intercept absorbs much of the latent variation, indicating that some releases revise many countries at once. This is consistent with releases silently correcting errors that were not officially documented.

5.2 Decomposing the Mechanism

Table 3 decomposes the democracy finding by examining which specific aspects of democratic governance are associated with data revisions. *Media Freedom* is positive ($p < 0.01$) for both outcomes: where governments refrain from directly or indirectly attempting to censor print or broadcast media, revisions are both more frequent and larger. *Freedom of Academic Exchange* has a positive and significant association with likelihood ($p < 0.01$), but not with magnitude. This asymmetry suggests that academic freedom enables the *decision* to revise, but it does not independently determine *how large* the revision is once the decision has been

made.

The association between democracy and revision behavior runs through press and academic freedom. As Model 1 shows, no alternative mechanism is significantly associated with the likelihood of revision. Still, the electoral competition and popular participation mechanisms influence the magnitude, as Model 2 shows. The share of population with *Suffrage* is positive and significant ($p < 0.01$), whereas the vote share of the largest opposition party in the most recent legislative election (*Opposition Strength*) is negative ($p < 0.01$). As the coefficient for *Political Corruption Index* indicates, more corrupt environments significantly suppress the size of revisions even when revisions do occur ($p < 0.01$), consistent with officials having incentives to minimize the visibility of corrections rather than acknowledge past errors fully. The remaining results are similar to those in Table 2.[19] *Public Administration* loses significance, likely because the democracy component indicators capture overlapping aspects of institutional quality that absorb the independent contribution of bureaucratic capacity.

Revisions are most likely when journalists, academics, and other experts can freedom demand data transparency and push for corrections. This finding speaks directly to the validity-reliability trade-off: the institutional conditions that produce more valid data (freedom of academic expression, independent expertise, and the resulting transparency pressures) also produce less reliable data. In contrast, pressure from voters and political rivals is insufficient to increase the likelihood of revision (though these factors do matter for the magnitude of revision).

Additional tests confirm that the democracy finding is not an artifact of measurement choices, sample composition, or model specification. Appendix D presents models with alternative measures of state capacity, economic development, EU membership, and international lending. The democracy finding remains robust throughout. Appendix E restricts the analy-

[19] One difference is that *Data Management Error* is excluded from Model 1 of Table 3 due to separation issues. Due to missing values in the democracy components, the models in Table 3 rely on smaller samples than the equivalent models in Table 2. Idiosyncratic errors are rare events: in a reduced sample, there are too few such observations to reliably estimate the *Data Management Error* coefficient in the logistic regression.

Table 3: Determinants of the Likelihood and Magnitude of Data Revisions: Which Aspects of Democracy Matter?

	Dependent Variable:	
	Revision	Abs. % Change
	(1)	(2)
Media Freedom, $y-1$	0.05^{***}	0.29^{***}
	(0.01)	(0.06)
Freedom of Academic Exchange, $y-1$	0.09^{***}	-0.16
	(0.02)	(0.10)
Suffrage, $y-1$	0.09	0.91^{**}
	(0.11)	(0.45)
Opposition Strength, $y-1$	-0.02	-0.57^{***}
	(0.05)	(0.20)
Turnover Frequency, Prev. 5 Yrs, $y-1$	0.00	0.05
	(0.01)	(0.06)
Political Corruption Index, $y-1$	0.05	-1.57^{***}
	(0.10)	(0.39)
Public Administration, $y-1$	0.03	0.05
	(0.02)	(0.09)
IMF Program, $y-1$	0.02	-0.09
	(0.02)	(0.08)
Financial Crisis, $y-1$	-0.01	-0.17^{**}
	(0.02)	(0.08)
Natural Disaster, $y-1$	0.06^{***}	-0.11
	(0.02)	(0.07)
Armed Conflict, $y-1$	0.04	-0.16
	(0.03)	(0.14)
SNA Change, v	0.37^{***}	1.29^{***}
	(0.03)	(0.11)
Data Management Error, v		11.14^{***}
		(1.53)
Log Difference, $v-y$	-0.99^{***}	-4.70^{***}
	(0.04)	(0.11)
Intercept	-2.73^{***}	0.22
	(0.58)	(1.06)
AIC	139065.27	391573.36
Log Likelihood	-69515.64	-195767.68
Observations	350529	58721
Number of Countries	166	166
Number of Reference Years	30	30
Number of Vintages	103	78
Variance: Country (Intercept)	0.24	2.92
Variance: Reference Year (Intercept)	0.03	12.84
Variance: Vintage (Intercept)	28.44	26.68

Notes: This table presents a logistic regression (Model 1) and a linear regression (Model 2) with random intercepts for country, year, and vintage. *Data Management Error* is excluded from Model 1 due to near-perfect separation. y is the reference year; v is the vintage. Standard errors in parentheses. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

sis to the main scheduled annual release, reducing noise from the increased update frequency in recent years. The likelihood finding holds, while the magnitude finding is weaker in this smaller sample. Appendix F replaces random effects with fixed effects; the Polyarchy finding survives for likelihood but is less consistent for magnitude, as expected given the efficiency costs of three-way fixed effects. Appendix G examines the direction of revisions: upward revisions are significantly more likely at higher Polyarchy scores, consistent with democracies publishing cautious initial estimates that are later adjusted upward as better information arrives.

5.3 The Timing of Democracy

My theory locates the decisive conditions at the data collection stage: democratic institutions make data transparent enough to be traced and corrected later. This implies that democracy at y should predict revision regardless of the regime in place at v, when the revision is actually made. Alternatively (or additionally), revisions might respond to contemporaneous pressure from a free press and independent experts *at the moment of correction*. If so, then democracy at v should matter. Table 4 tests for this possibility. Models 1 and 3 include democracy at both y and v. *Polyarchy* at y is positively and significantly associated with both the likelihood and the magnitude of revision, whereas the coefficient for *Polyarchy* at v is smaller and less stable. However, these separate coefficients cannot be cleanly identified, as democracy values at y and v are highly correlated ($r = 0.88$). Therefore, I turn to the interaction (Models 2 and 4) and its marginal effects (Figure 4) to understand how the two stages relate.

Polyarchy at y is positively and significantly associated with the likelihood of revision at every level of *Polyarchy* at v, with no significant interaction (Model 2). GDP collected under more democratic conditions is more likely to be revised, regardless of the country's regime at the time of revision. The magnitude of revisions behaves differently (Model 4 and Figure 4). The association between democracy at y and revision size is indistinguishable from zero

Table 4: Determinants of the Likelihood and Magnitude of Data Revisions: The Timing of Democracy

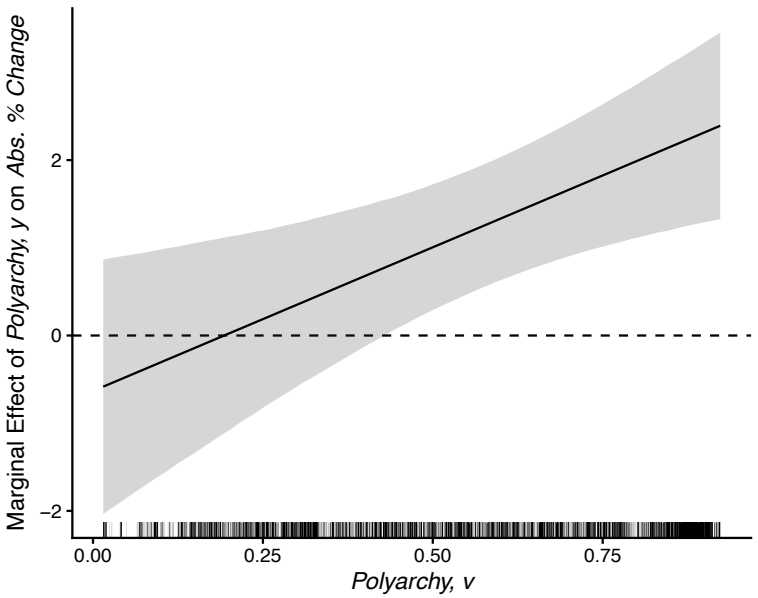
	Dependent Variable:			
	Revision	Revision	Abs. % Change	Abs. % Change
	(1)	(2)	(3)	(4)
Polyarchy, $y-1$	0.65^{***}	0.88^{***}	1.24^{***}	-0.63
	(0.09)	(0.19)	(0.36)	(0.76)
Polyarchy, $v-1$	-0.35^{***}	-0.17	0.26	-1.04^{*}
	(0.10)	(0.16)	(0.38)	(0.60)
Polyarchy, $y-1 \times$ Polyarchy, $v-1$		-0.41		3.28^{***}
		(0.30)		(1.17)
Public Administration, $y-1$	0.04^{**}	0.04^{**}	0.12	0.11
	(0.02)	(0.02)	(0.08)	(0.08)
IMF Program, $y-1$	0.02	0.02	-0.14^{*}	-0.12
	(0.02)	(0.02)	(0.08)	(0.08)
Financial Crisis, $y-1$	0.00	0.00	-0.16^{**}	-0.16^{**}
	(0.02)	(0.02)	(0.08)	(0.08)
Natural Disaster, $y-1$	0.05^{***}	0.05^{***}	-0.11	-0.11
	(0.02)	(0.02)	(0.07)	(0.07)
Armed Conflict, $y-1$	0.05	0.04	-0.23^{*}	-0.21
	(0.03)	(0.03)	(0.13)	(0.13)
SNA Change, v	0.36^{***}	0.36^{***}	1.26^{***}	1.25^{***}
	(0.03)	(0.03)	(0.11)	(0.11)
Data Management Error, v	6.28^{***}	6.29^{***}	11.51^{***}	11.52^{***}
	(1.15)	(1.15)	(1.49)	(1.49)
Log Difference, $v-y$	-1.02^{***}	-1.02^{***}	-4.66^{***}	-4.65^{***}
	(0.03)	(0.03)	(0.10)	(0.10)
Intercept	-2.35^{***}	-2.42^{***}	-0.36	0.19
	(0.56)	(0.56)	(0.93)	(0.95)
AIC	148365.83	148365.95	420301.27	420295.51
Log Likelihood	-74168.92	-74167.98	-210135.64	-210131.76
Observations	375247	375247	63050	63050
Number of Countries	169	169	169	169
Number of Reference Years	31	31	31	31
Number of Vintages	103	103	78	78
Variance: Country (Intercept)	0.24	0.25	3.06	2.81
Variance: Reference Year (Intercept)	0.05	0.05	12.61	12.61
Variance: Vintage (Intercept)	27.88	27.89	26.18	26.25

Notes: This table presents two logistic regressions (Models 1–2) and two linear regressions (Models 3–4) with random intercepts for country, year, and vintage. y is the reference year; v is the vintage. Standard errors in parentheses. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

when the vintage regime is autocratic (a marginal effect of -0.08 at the 10th percentile of vintage-year democracy), but large and significant when it is democratic (2.24, $p < 0.01$, at the 90th percentile).

In sum, *whether* a democracy revises depends on the conditions under which the data were

Figure 4: Marginal Effect of Reference-Year Democracy on Revision Magnitude Across Vintage-Year Democracy



This figure presents the marginal effect of democracy at the reference year (*Polyarchy, y*) on revision magnitude across the range of vintage-year democracy (*Polyarchy, v*), based on Model 4 of Table 4. The effect is indistinguishable from zero when the vintage regime is autocratic and increases as it becomes more democratic. The shaded band is the 95% confidence interval; marks at the base show the distribution of vintage-year democracy among revised observations.

originally produced, as my theory expects. *How much* a democracy revises also depends on contemporaneous scrutiny: large corrections require experts to be watching when the revision is made, not only when the data were first reported. In Appendix D, I report subgroup models estimated separately within autocratic-vintage and democratic-vintage country-years, which corroborate the results reported here.

6 Discussion and Conclusion

This study shows that, conditional on state capacity, revisions are more likely and larger for GDP data collected by democracies. The finding is robust across three alternative measures of regime type: *Polyarchy*, *Polity 2*, and *Freedom House*. Beyond GDP data, I expect these predictions to hold for other macroeconomic indicators, like FDI, trade, inflation, and

unemployment, all of which are regularly revised. Beyond WDI data, the argument likely travels to other data sources. Since the Penn World Table (PWT) and the Maddison Project Database are not updated as frequently, they do not allow for as systematic a test.[20] Still, future research can validate the generalizability of my results by exploring variations in the frequency and magnitude of revisions across other indicators and data sources.

These results have four key implications. First, they add nuance to existing evidence that democracies produce higher-validity data. Democratically elected leaders might be less likely to lie about growth rates or COVID-19 deaths (Martínez, 2022; Adiguzel et al., 2020), yet they face greater incentives to scrutinize and revise information. As a result, democracies produce less reliable data: their numbers are more likely to change over time. Even good-faith revisions can create a perception of mismanagement, an issue that is politically costlier for democracies (which are held to higher transparency standards than autocracies). In contrast, autocrats might collect biased data and publish imprecise estimates, but they rarely revise these estimates after publication. For different reasons, all data are flawed.

Second, transparency in revision processes must be accompanied by public communication strategies that frame revisions as a sign of statistical rigor. The backlash against BLS employment data updates in 2024 and 2025 illustrates how, in contexts of low data literacy and widespread misinformation, revisions can erode confidence in official statistics, undermine the government's authority as the main producer of macroeconomic data, and drive individuals toward private data providers.[21] Breaking this cycle requires a broader public understanding of why reliable data and valid data are different, and why pursuing one can come at the expense of the other.

Third, the initial data-collection environment matters more than subsequent adjustments. Today's improvements in statistical capacity do not automatically fix past data, and the sequencing of data-quality investments matters as much as their magnitude.

[20] For example, between 1994 and 2024, there have been eleven PWT releases and five releases of the Maddison Project Database, compared to over 100 WDI vintages available in the WDI Database Archives.

[21] Geoffrey Morgan and Matthew Griffin. "Wall Street Leaning Harder on Private Data After Trump BLS Spat." *Bloomberg*. 23 August 2025.

Finally, my results extend work on the discrepancies between WDI and PWT vintages, or between different vintages of the same data source (Goes, 2023; Johnson et al., 2013; Croushore and Stark, 2003). While much of the literature has focused on non-democracies (Goes, 2023) or African states (Johnson et al., 2013), my results suggest that industrial democracies also warrant scrutiny: their data may be more valid but less stable than assumed. Researchers using data from democracies trade reliability for validity, whereas those using data from autocracies face the opposite trade-off. Neither choice is neutral, and researchers should be explicit about which dimension their analysis depends on most.

In practice, testing a result's sensitivity to different vintages is rarely feasible, since multiple variables are vintaged and scattered across sources. Still, researchers should report the vintage they use and, where possible, prefer recent ones. They should avoid patching data across vintages to maximize coverage, as this introduces measurement error correlated with regime type. Because this instability is concentrated among democracies, such caution matters most where one might least expect it: democracy-heavy samples. As McMann et al. (2022) show, discussions about data quality often focus on validity and neglect reliability. This study shows that the two are bound together by the politics of data production — and that understanding one requires understanding the other.

Declaration of Competing Interests

The author declares no competing interests.

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Appendix for
The Price of Accountability: Democratic Institutions and the (In)Stability of Economic Data

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A Countries Included in the Analysis

The following countries are included in the main analysis; some drop out in robustness checks due to more limited data coverage.

Afghanistan, Albania, Algeria, Angola, Argentina, Armenia, Australia, Austria, Azerbaijan, Bahrain, Bangladesh, Barbados, Belarus, Belgium, Benin, Bhutan, Bolivia, Bosnia and Herzegovina, Botswana, Brazil, Bulgaria, Burkina Faso, Burundi, Cambodia, Cameroon, Canada, Cape Verde, Central African Republic, Chad, Chile, China, Colombia, Comoros, Costa Rica, Côte d'Ivoire, Croatia, Cuba, Cyprus, Czechia, Democratic Republic of the Congo, Denmark, Djibouti, Dominican Republic, Ecuador, Egypt, El Salvador, Equatorial Guinea, Eritrea, Estonia, Eswatini, Ethiopia, Fiji, Finland, France, Gabon, Gambia, Georgia, Germany, Ghana, Greece, Guatemala, Guinea, Guinea-Bissau, Guyana, Haiti, Honduras, Hungary, Iceland, India, Indonesia, Iran, Iraq, Ireland, Israel, Italy, Jamaica, Japan, Jordan, Kazakhstan, Kenya, Kuwait, Kyrgyzstan, Laos, Latvia, Lebanon, Lesotho, Liberia, Libya, Lithuania, Luxembourg, Madagascar, Malawi, Malaysia, Maldives, Mali, Malta, Mauritania, Mauritius, Mexico, Moldova, Mongolia, Montenegro, Morocco, Mozambique, Myanmar, Namibia, Nepal, Netherlands, New Zealand, Nicaragua, Niger, Nigeria, North Macedonia, Norway, Oman, Pakistan, Panama, Papua New Guinea, Paraguay, Peru, Philippines, Poland, Portugal, Qatar, Republic of the Congo, Romania, Russia, Rwanda, São Tomé and Príncipe, Saudi Arabia, Senegal, Serbia, Sierra Leone, Singapore, Slovakia, Slovenia, Solomon Islands, South Africa, South Korea, South Sudan, Spain, Sri Lanka, Sudan, Suriname, Sweden, Switzerland, Syria, Tajikistan, Tanzania, Thailand, Timor-Leste, Togo, Trinidad and Tobago, Tunisia, Turkey, Turkmenistan, Uganda, Ukraine, United Arab Emirates, United Kingdom, United States, Uruguay, Uzbekistan, Vanuatu, Venezuela, Vietnam, Yemen, Zambia, Zimbabwe.

B Variables Included in the Analysis

Reference year predictors (y) are lagged by one year. In the following table, the availability column lists the reference years for which the lagged versions are used. Vintage year predictors enter contemporaneously (v). Some variables (such as *Information Capacity* and *State Capacity*) are available for shorter periods, as indicated.

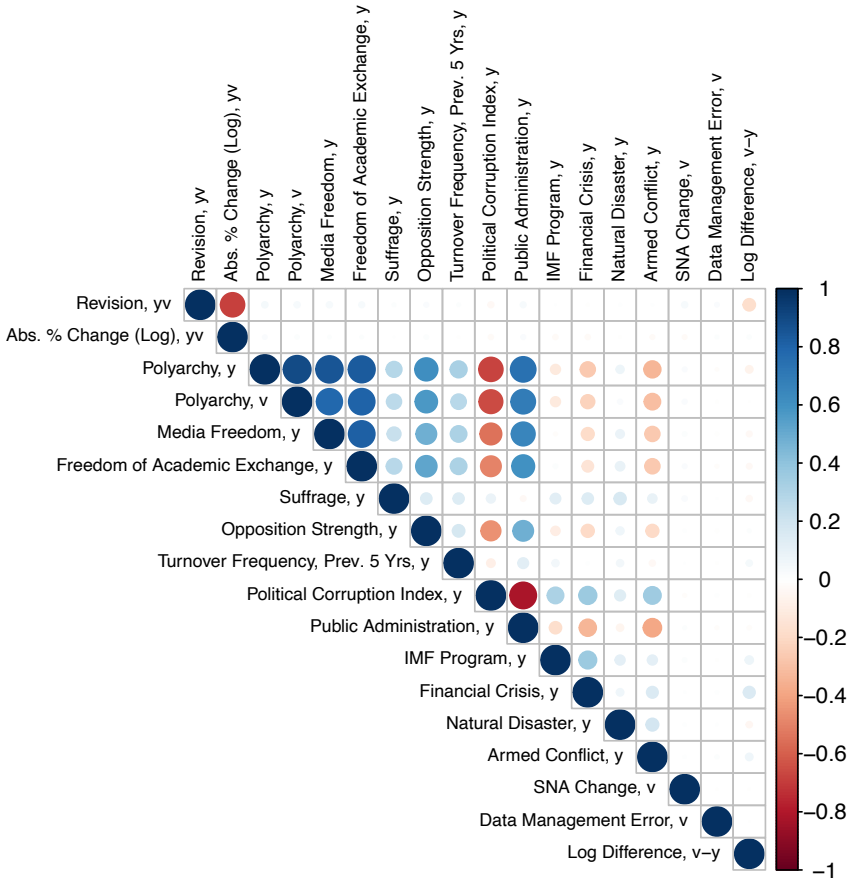
Table B.1: Variables Included in the Analysis

Variable	Description	Coverage	Source
Absolute % Change (Log), yv	Absolute value of the percentage change in GDP between two consecutive WDI releases for the same country-year pair, logged. Defined only when a revision occurs and neither value is missing	1990–2021	WDI
Armed Conflict, $y-1$	Was there an armed conflict? Yes = 1	1990–2020	Gleditsch et al. (2002)
Bureaucratic Quality, $y-1$	To what extent does the country's bureaucracy have the strength and expertise to govern without drastic changes in policy or interruptions in government services? Low = 0, High = 4	1990–2020	The PRS Group (2022)
Data Management Error, v	Coded 1 for the following observations: Armenia, 1992, April 1998 WDI; China, 2007 and 2008, May 2009 WDI; Democratic Republic of the Congo, 1990, July 2016 to April 2017 WDI; Estonia, 1995, April 1997 WDI; Georgia, 1990, April 1999 and April 2000 WDI; Myanmar, all years, December 2021 WDI	1994–2021	Own coding, based on World Bank (2023)
Democracy, $v-1$	Dichotomous indicator derived from the V-Dem Regimes of the World typology: 1 for electoral and liberal democracies, 0 for closed and electoral autocracies	1990–2021	Coppedge et al. (2023); Lührmann et al. (2018)
EU Membership, $y-1$	Was the country a member of the EU that year? Yes = 1	1990–2020	Own coding, based on accession dates
Financial Crisis, $y-1$	Was there a banking, currency, or debt crisis? Yes = 1	1990–2020	Nguyen et al. (2022)
Freedom House, $y-1$	Combined measure of political rights and civil liberties, originally from 1 (most free) to 7 (least free); scale reversed so that higher values indicate more freedom	1990–2020	Freedom House (2026)
Freedom of Academic Exchange, $y-1$	Are scholars free to exchange and communicate research findings, including through media engagement and public lectures? (not = 0, weakly = 1, somewhat = 2, mostly = 3, fully = 4). Official V-Dem variable name: *Freedom of Academic Exchange and Dissemination*	1990–2020	Coppedge et al. (2023)
IMF Commitment (Log), $y-1$	Total IMF commitments (GRA purchases and PRGT disbursements), logged	1990–2020	IMF (2026)
IMF Disbursement (Log), $y-1$	Total IMF loan disbursements, logged	1990–2020	IMF (2026)
IMF Program, $y-1$	Was there an IMF program? Yes = 1	1990–2020	Kentikelenis and Stubbs (2023)
Income Group, $y-1$	World Bank historical income classification: low income = 1, lower middle income = 2, upper middle income = 3, and high income = 4.	1990–2020	World Bank (2026a)

Information Capacity, $y-1$	Latent measure of state information capacity, estimated via item response theory	1990–2012	Brambor et al. (2020)
Log Difference, $v-y$	Number of years elapsed between the vintage and the reference year, logged	1990–2020	WDI
Media Freedom, $y-1$	Does the government refrain from censoring political information in the media? (not = 0, weakly = 1, somewhat = 2, mostly = 3, fully = 4). Official V-Dem variable name: *Government Censorship Effort — Media*	1990–2020	Coppedge et al. (2023)
Natural Disaster, $y-1$	Was there a biological (epidemic), climatological (drought, wildfire), meteorological (storm, extreme temperature), hydrological (flood, landslide), or geophysical (earthquake, volcanic activity) disaster? Yes = 1	1990–2020	Centre for Research on the Epidemiology of Disasters (2020)
Opposition Strength, $y-1$	Vote share of the largest opposition party in the most recent legislative election	1990–2020	Cruz et al. (2021)
Political Corruption Index, $y-1$	Average levels of public sector, executive, legislative, and judicial corruption, from 0 (less corrupt) to 1 (more corrupt)	1990–2020	Coppedge et al. (2023)
Polity 2, $y-1$	Revised combined Polity score, from −10 (hereditary monarchy) to +10 (consolidated democracy)	1990–2020	Marshall and Gurr (2020)
Polyarchy, $y-1$ and $v-1$	Electoral democracy index, from 0 to 1	1990–2021	Coppedge et al. (2023)
Public Administration, $y-1$	Are public officials rigorous and impartial in the performance of their duties? (no = 0, weakly = 1, modestly = 2, mostly = 3, fully = 4). Official V-Dem variable name: *Rigorous and Impartial Public Administration*	1990–2020	Coppedge et al. (2023)
Revision, yv	Did the GDP value change between two consecutive WDI releases for the same country-year pair? Yes = 1. Coded NA for the first vintage of each country-year pair	1990–2020	WDI
SNA Change, v	Was the System of National Accounts in use updated this year? Yes = 1	1994–2021	UN National Accounts Statistics, complemented by WDI Metadata and IMF International Financial Statistics
State Capacity, $y-1$	Latent measure of state capacity combining 21 indicators of extractive, coercive, statistical, informational, and administrative capacity	1990–2015	Hanson and Sigman (2021)
Suffrage, $y-1$	Share of adult citizens that have the legal right to vote in national elections	1990–2020	Coppedge et al. (2023)
Turnover Frequency, $y-1$	Number of changes in party control of the executive in the preceding five years	1990–2020	Cruz et al. (2021)

Upward Revision, yv	Among revisions where neither value is missing, did the GDP value increase? Yes = 1. Coded NA when no revision occurred or a value is missing	1990–2021	WDI
WB Commitment (Log), $y-1$	Total World Bank (IBRD and IDA) commitments, logged	1990–2020	World Bank (2026b), World Bank (2026c)
WB Disbursement (Log), $y-1$	Total World Bank (IBRD and IDA) disbursements, logged	1990–2020	World Bank (2026b), World Bank (2026c)

Figure B.1: Correlation Plot for Variables Used in the Main Text



This figure shows the correlation plot for all variables included in the main text. To generate this figure, *Abs.% Change (Log)* was coded as zero for all instances where no revision occurred (*Revision = 0*) or where the concept of revision is undefined (*Revision = NA*), in order to include all observations in the correlation matrix. In the actual analysis, *Abs.% Change (Log)* is undefined for non-revisions and the magnitude models are estimated only on observations where a revision occurred and neither GDP value is missing.

Table B.2: Summary Statistics

Variable	N	Mean	Std. Dev.	Min	Pctl. 25	Pctl. 75	Max
Absolute % Change (Log), yv	64822	-9.416	9.057	-30.02	-17.89	-0.9101	9.769
Armed Conflict, $y-1$	405089						
... 0	298272	73.63%					
... 1	106817	26.37%					
Bureaucratic Quality, $y-1$	316092	2.17	1.152	0	1	3	4
Data Management Error, v	418141						
... 0	418090	99.99%					
... 1	51	0.01%					
Democracy, $v-1$	410217						
... 0	192402	46.9%					
... 1	217815	53.1%					
EU Member, $y-1$	405089						
... 0	359877	88.84%					
... 1	45212	11.16%					
Financial Crisis, $y-1$	405089						
... 0	237749	58.69%					
... 1	167340	41.31%					
Freedom House, $y-1$	396723	-3.669	1.838	-7	-5	-2	-1
Freedom of Academic Exchange, $y-1$	378471	2.858	1.083	0	2	4	4
IMF Commitment (Log), $y-1$	405089	2.225	6.091	0	0	0	24.86
IMF Disbursement (Log), $y-1$	405089	4.591	7.683	0	0	14.83	23.73
IMF Program, $y-1$	405089						
... 0	271168	66.94%					
... 1	133921	33.06%					
Income Group, $y-1$	403945	2.292	1.121	1	1	3	4
Information Capacity, $y-1$	142520	0.6459	0.1493	0.2387	0.6051	0.7945	1
Log Difference, $v-y$	418141	2.245	0.9377	-1.099	1.792	2.944	3.434
Media Freedom, $y-1$	385487	2.355	1.409	0	1	3	4
Natural Disaster, $y-1$	405089						
... 0	156987	38.75%					
... 1	248102	61.25%					
Opposition Strength, $y-1$	399810	0.2001	0.222	0	0	0.415	0.995
Political Corruption Index, $y-1$	398683	0.5109	0.3023	0.002	0.207	0.785	0.967
Polity 2, $y-1$	377903	2.846	6.722	-10	-4	9	10
Polyarchy, $v-1$	410217	0.5262	0.259	0.015	0.296	0.766	0.923
Polyarchy, $y-1$	400255	0.497	0.2742	0.013	0.249	0.762	0.923
Public Administration, $y-1$	400255	2.206	1.094	0	1	3	4
Revision, yv	385602						
... 0	320233	83.05%					
... 1	65369	16.95%					
SNA Change, v	413035						
... 0	388396	94.03%					
... 1	24639	5.97%					
State Capacity, $y-1$	369347	0.4132	0.9589	-2.029	-0.2548	0.9487	2.964
Suffrage, $y-1$	400255	0.9701	0.1615	0	1	1	1
Turnover Frequency, Prev. 5 Yrs, $y-1$	399572	0.4904	0.6054	0	0	1	3
Upward Revision, yv	64822						
... 0	28783	44.4%					
... 1	36039	55.6%					
WB Commitment (Log), $y-1$	405089	9.437	9.324	0	0	18.48	22.58
WB Disbursement (Log), $y-1$	405089	9.693	9.717	0	0	19.15	26.16

C Additional Descriptive Information

Unlike latent concepts like election integrity, state capacity, or democratic consolidation, which rely on expert coding that might be ideologically biased or improperly aggregated (Giannone, 2010; Martínez i Coma and van Ham, 2015; Hanson and Sigman, 2021), macroeconomic indicators purportedly measure observable, neutral, "objective" outputs. Still, these indicators are often accused of oversimplifying abstract

concepts like wealth, inequality, or unemployment (Mügge, 2022). For example, GDP excludes unpaid household services, which are disproportionately performed by women (DeRock, 2021). Sometimes different data sources oversimplify the world in different ways. As a result, the two most common data sources in political science and economics — the WDI and the Penn World Table, respectively (Goes, 2023; Johnson et al., 2013) — provide estimates that can differ by over 25% (Ram and Ural, 2014). Exporters and importers record the same bilateral trade flows differently (Linsi et al., 2023), and a comparison of export data from two sources — the IMF and the UN Commodity Trade Statistics — concludes that oftentimes the data are not even correlated (Amin Gutiérrez de Piñeres, 2006, 35). Climate aid (Michaelowa and Michaelowa, 2011; Weikmans and Roberts, 2019) and FDI (Kerner, 2014) suffer from similar discrepancies. Even within one country, different agencies might compete for data collection, presenting contradictory results (Pellechio and Cady, 2006). The US is among the few countries without a centralized NSO; its federal statistical system consists of over 100 statistical agencies, units, and programs that might provide conflicting information. And even an oversimplified world is difficult to measure.

This section provides more detailed information about the challenges of measuring valid macroeconomic data. First, I provide detailed examples of prominent information-based and structural revisions, illustrating the range of magnitude and direction of revisions in the dataset. Then, I discuss the threats to validity that produce error-driven revisions — the idiosyncratic and capacity-related problems that make preliminary data incorrect enough to require correction in the first place.

C.1 Information-Based and Structural Revisions

About 39.8% of all economies surveyed by the IMF in 2020 first released annual GDP data within 90 days of the reference period, whereas 54.4% did so within 91 to 365 days, with no available information for the remaining 5.8% (Baer et al., 2022, 17). Many of these economies later revised their preliminary data, as recommended by international organizations. This section provides additional information about prominent data revisions. I begin by providing examples of what Croushore and Stark (2003) deems information-based revisions, which are related to the "discovery" of new data. In these cases, countries revise their data to incorporate newly available information that allows for more precise measurement.

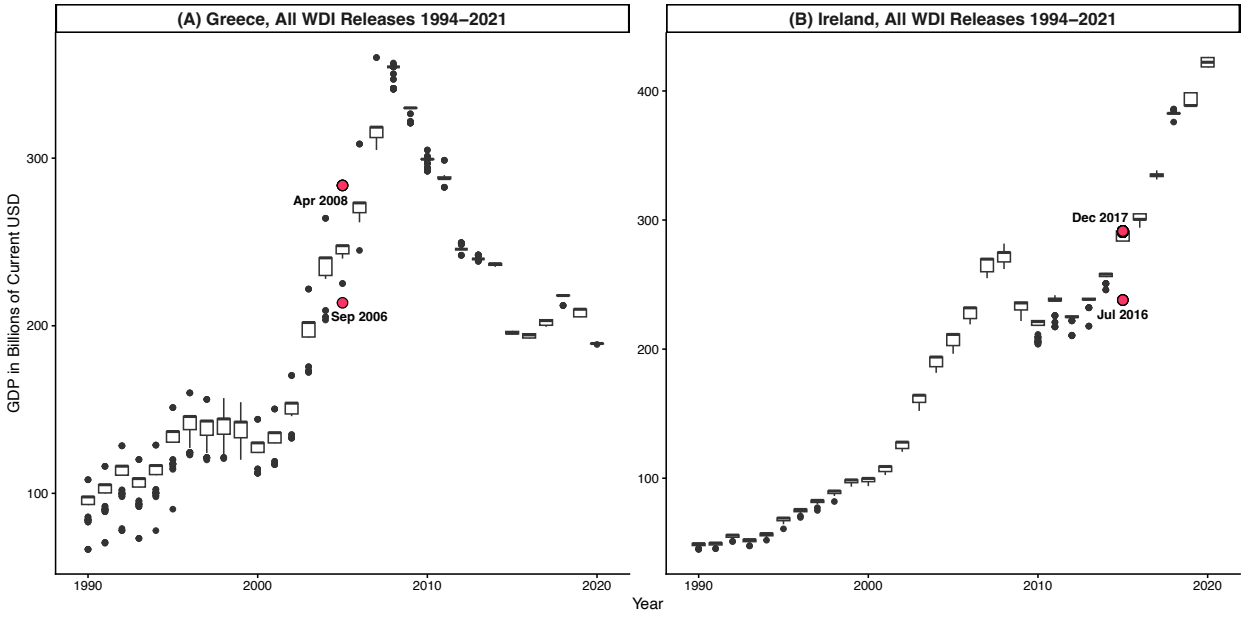
In 2009, Prime Minister George Papandreou came to power in Greece and requested help from Eurostat and the International Monetary Fund (IMF) to revise the country's finances.[1] As these revisions quickly established, previous administrations had overestimated tax revenues, disregarded social security and military expenses, and engaged in creative accounting to hide government liabilities. Between April and October 2009,

[1] In fact, information-based revisions are often tied to international actors; for example, subscribing to the IMF's Special Data Dissemination Standard (SDDS) can increase transparency (Vadlamannati et al., 2018).

the planned deficit ratio for 2009 was revised from 3.7 to 12.5% of the GDP (European Commission, 2010), a number that — according to Eurostat — was still far below the truth (Aragão and Linsi, 2022). Following these revisions, credit rating agencies downgraded Greece, which requested multiple IMF and EU loans to avoid default.

A revision in Ireland caused similar controversy. From 2015 to 2016, the Irish Central Statistics Office reported a GDP growth of 26.3%. Citing statistical confidentiality rules, authorities initially refused to release detailed data, instead devising a new statistic — the Modified Gross National Income (GNI*) — to remove "globalization-related" distortions. Ireland's "leprechaun economics" became a source of ridicule, and the country was blacklisted as a tax haven. Years later, economists discovered that these distortions had been driven by Apple's decision to onshore intellectual property assets to Ireland in 2015 (Polyak, 2023). Figure C.1 illustrates the cases of Greece and Ireland. Between the September 2006 and April 2008 WDI releases, Greece's GDP for 2005 increased by nearly 33%, from 213.7 billion to 283.7 billion. Between the July 2016 and the December 2017 WDI, Ireland's 2015 GDP figures increased by 22.6%.

Figure C.1: Current GDP of Greece and Ireland, 1990–2020



These boxplots present the distribution of current GDP estimates for (A) Greece and (B) Ireland from 1990 to 2020, using data drawn from the 104 WDI releases from April 1994 to December 2021. The estimate reported for Greece in 2005 is 70 billion dollars (32.78%) larger in the April 2008 WDI than in the April 2007 WDI. The estimate reported for Ireland in 2015 is 53.8 billion dollars (22.6%) larger in the December 2017 WDI than in the July 2016 WDI.

Beyond Europe, the Ghana Statistical Service released new GDP estimates in 2010, with support from the Danish International Development Agency and the IMF. After upgrading from the 1968 to the 1993 System of National Accounts (SNA), updating the base year, and disaggregating data by economic sector, it concluded

that the country's GDP was 60.3% larger than previously thought (Jerven and Ebo Duncan, 2012). Kenya, Nigeria, and others similarly reported GDP increases after incorporating new information from informal activities (African Development Bank, 2013). In the following years, the World Bank upgraded all three countries from low income to lower middle income economy. This shift was associated with less generous lending terms, as countries with per capita incomes above a certain threshold lose access to concessional lending (Kerner et al., 2017).

Besides information-based revisions, countries may also conduct structural revisions, changing definitions, base years, or aggregation methods. For example, the 1993 SNA introduced the concept of imputed rent, which significantly altered the definition and calculation of GDP (Olinto Ramos et al., 2008). Nicaragua revised its national accounts in 2003, changing the base year from 1980 to 1994 and implementing the 1993 SNA; as a result, the country's current GDP for the year 2000 increased by 70% (Olinto Ramos et al., 2008, 9). As mentioned in the main text, EU countries revised their GDP definition to include drug trafficking and prostitution in 2014; as a result, the Italian and British economies increased by 4% each (Coyle, 2014, 110). For quarterly data, at least, information-based revisions are most common up to one year after the initial data release, with structural revisions increasing in frequency as time goes by (Croushore and Stark, 2003).

C.2 Sources of Inaccurate Data

Error-driven revisions arise when preliminary data are inaccurate from the outset. Three factors often make preliminary data incorrect enough to require correction: low statistical capacity, political incentives to misreport, and idiosyncratic data management errors.

The first source of inaccuracy is low statistical capacity. A 2005 survey by the United Nations Economic Commission for Africa (2005) found that some NSOs in the continent had as few as three national accountants. A 2023 survey of 14 NSOs, conducted by the Inter-American Development Bank, found that only half of the employees working with statistical analysis displayed basic competence in probability, descriptive statistics, survey sampling, and arithmetic (Mejía Guerra et al., 2023, 14). Many lacked the expertise to report data consistent with the System of National Accounts (SNA), a global standardization framework developed by the International Comparison Program (ICP) to enable cross-country comparisons. Communist countries did not begin to adopt the SNA until 1993, and North Korea still uses a Marxism-inspired alternative, the Material Product System (Herrera 2010, 23n8; van Heijster and DeRock 2022, 84n1), rendering its data incomparable to other countries.

Even countries that follow the SNA might struggle with difficult-to-measure concepts like imputed rent,

thus underestimating household final consumption expenditure — an important component of GDP (Olinto Ramos et al., 2008). Moreover, it is difficult to quantify the size of the informal economy, which accounts for up to 44% of the GDP in the developing world (Coyle, 2014, 110). Population figures tend to be extrapolated from the last census and can grow progressively inaccurate over time; Lebanon, for instance, last conducted a census in 1932 (Devarajan, 2013). Some countries struggle with overly ambitious data-collection efforts: Chile's 2012 census, which spanned three months, failed to account for nearly 10% of the population and was replaced by a single-day census in 2017.

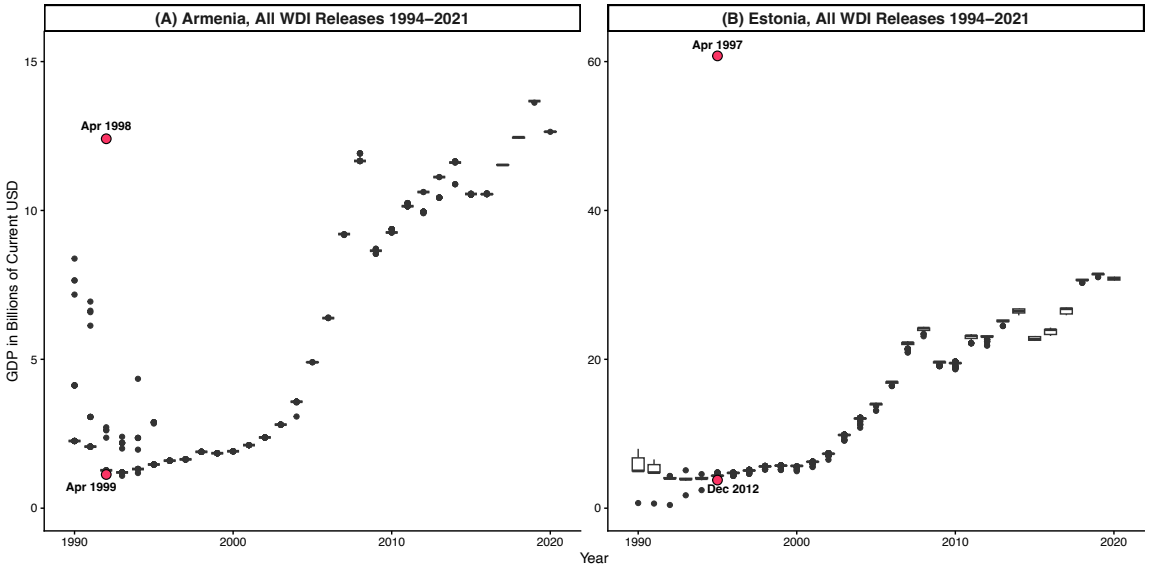
The second source of inaccuracy relates to political incentives. In federations like Nigeria, states inflate population figures to receive higher fiscal transfers from the federal government (Devarajan, 2013). Aid-dependent countries systematically understate their finances to appear poor and attract more aid (Kerner et al., 2017). Even industrialized democracies overstate how much climate aid they provide, particularly when domestic constituencies value environmental objectives (Michaelowa and Michaelowa, 2011), and misrepresent public finance statistics to abide by the rules of the European Union, as Greece did (Alt et al., 2014).

The final source of inaccuracy relates to idiosyncratic data management errors. These cannot be predicted systematically but are fairly easy to identify. Some trivial errors include: in 2006, "the country names for Burundi and Cameroon are in reverse order;" in 2008, "columns are incorrectly labeled as 1990; data are for 1995;" and in 2010, "an error for Zimbabwe's data" meant that several indicators, including GDP, "should be presented as not available for all years in the WDI database" (see World Bank 2023 for errata). Below, I provide concrete examples.

Figure C.2 presents the GDP of two former Soviet republics, Armenia and Estonia (both of which gained independence in 1991). According to the April 1998 WDI, Armenia had a GDP of 12.4 billion in 1992 — a number over four times as large as what any other WDI release reports. According to the April 1997 WDI, Estonia had a GDP of 60.8 billion in 1995 — a number at least 13 times as large as what other releases report. Since these extreme values only appear once, I assume they are the product of a data management error corrected in subsequent vintages.

Figure C.3 presents the GDP of two war-plagued countries in the Middle East, Iraq and Syria, and a newly independent country, Timor-Leste. Iraq in 1990 is an interesting case: this country-year pair first enters the WDI in September 1995 and takes the value of 48.66 billion until September 2005, at which point it ceases to be included. It reappears in the December 2014 WDI, at which point it is reported to be nearly four times as large: 179.91 billion. Syria's GDP from 1990 to 2010 is considerably larger in the December 2021 WDI than in other vintages. As of 2024, these values have not been revised; they are the most up-to-date values. The World Bank (2023) warns that *some* of its data series for Syria "should not

Figure C.2: Current GDP of Armenia and Estonia, 1990–2020



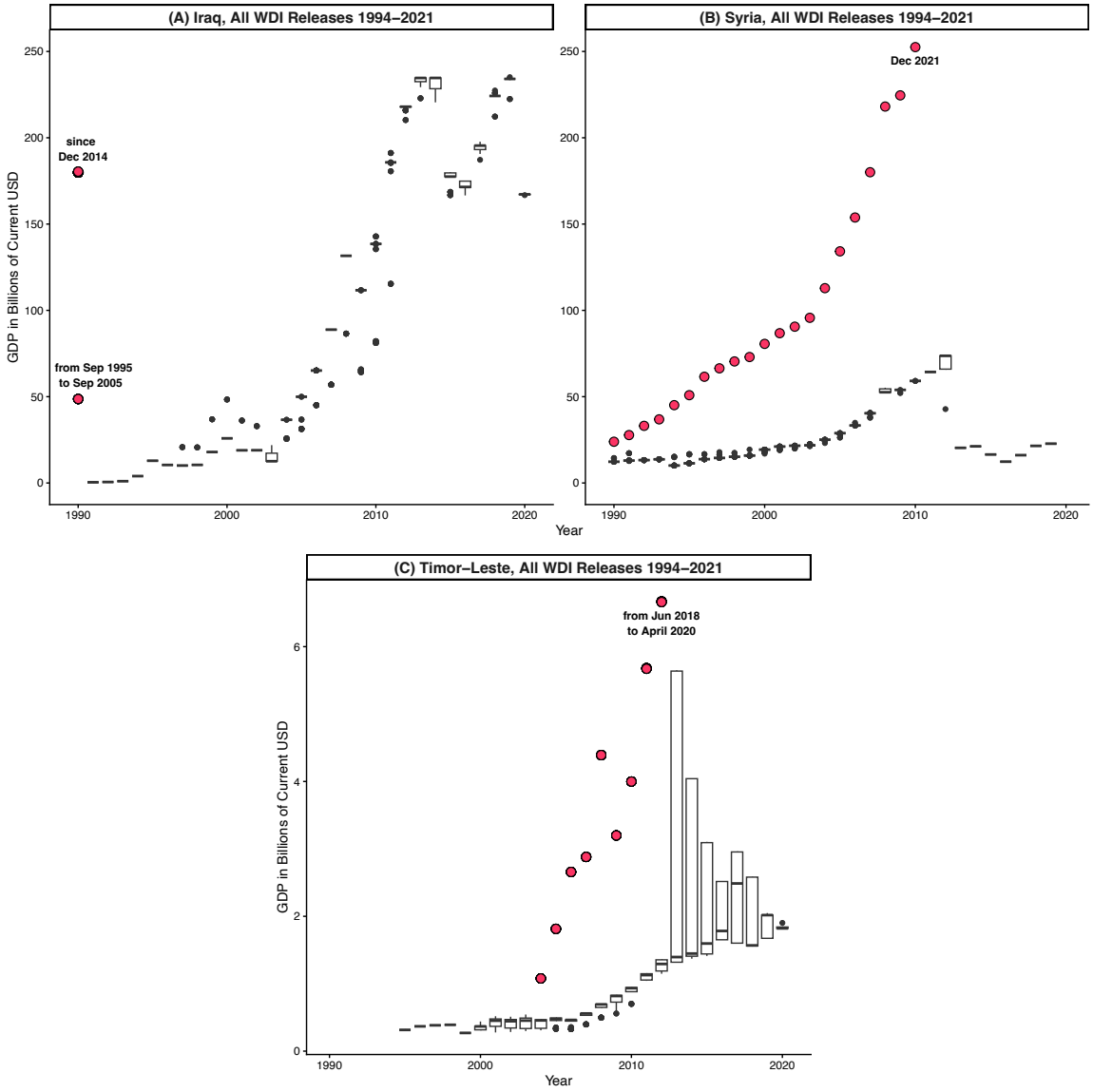
These boxplots present the distribution of current GDP estimates for (A) Armenia and (B) Estonia, from 1990 to 2020, using data drawn from the 104 WDI releases from April 1994 to December 2021.

be used for analysis or assessment" because they "are incomplete and based on estimates," but it does not specify *which* data series are affected by these issues.

As Figure C.3 further shows, all vintages report a GDP between 1.1 and 1.2 billion for Timor-Leste in 2012, with the exception of 16 vintages between June 2018 and April 2020 that report a number six times larger. Indeed, these 16 vintages provide exceptionally large values for Timor-Leste for all years from 2004 until 2012.

Iraq, Syria, and Timor-Leste's unusual values appear in multiple vintages. They are not specifically listed by the World Bank in its Data Updates and Errata website. In April 2022, I raised some of these issues to the WDI team via e-mail; three months later, a member of the World Bank's Development Data Group responded that they were related to the timing of the IT team's periodical maintenance and suggested I clear my browser cache, but doing so did not rectify these issues, and the WDI team did not respond to my follow-up inquiries about these specific observations. Thus, I do not code them as an error. In intentionally setting a high bar for an error, I likely underestimate the prevalence of such errors.

Figure C.3: Current GDP of Iraq, Syria, and Timor-Leste, 1990–2020



These boxplots present the distribution of current GDP estimates for (A) Iraq, (B) Syria, and (C) Timor-Leste from 1990 to 2020, using data drawn from the 104 WDI releases from April 1994 to December 2021.

D Alternative and Additional Controls

D.1 Alternative Measures of State Capacity

Table D.1 replaces *Public Administration* with three alternative measures of state capacity: the Hanson and Sigman (2021) state capacity index (Models 1 and 4), the ICRG measure of *Bureaucratic Quality* (Models 2 and 5), and the Brambor et al. (2020) information capacity index (Models 3 and 6). The democracy finding is robust across all six models: *Polyarchy* is positive and highly significant for both likelihood and

Table D.1: Determinants of the Likelihood and Magnitude of Data Revisions: Alternative Measures of State Capacity

	Dependent Variable:					
	Revision	Revision	Revision	Abs. % Change	Abs. % Change	Abs. % Change
	(1)	(2)	(3)	(4)	(5)	(6)
Polyarchy, $y-1$	0.64***	0.69***	0.28*	1.38***	1.55***	2.19***
	(0.08)	(0.09)	(0.17)	(0.33)	(0.36)	(0.57)
State Capacity, $y-1$	0.03			0.09		
	(0.03)			(0.13)		
Bureaucratic Quality, $y-1$		0.07***			0.12*	
		(0.02)			(0.07)	
Information Capacity, $y-1$			0.20			−3.14***
			(0.17)			(0.66)
IMF Program, $y-1$	0.03	0.02	0.01	−0.14*	−0.06	0.07
	(0.02)	(0.02)	(0.03)	(0.08)	(0.09)	(0.13)
Financial Crisis, $y-1$	−0.01	−0.00	−0.06*	−0.18**	−0.19**	−0.16
	(0.02)	(0.02)	(0.03)	(0.08)	(0.09)	(0.13)
Natural Disaster, $y-1$	0.05***	0.06***	0.03	−0.11	−0.11	−0.09
	(0.02)	(0.02)	(0.03)	(0.07)	(0.08)	(0.13)
Armed Conflict, $y-1$	0.04	0.03	−0.02	−0.27**	−0.21	−0.02
	(0.03)	(0.04)	(0.06)	(0.14)	(0.15)	(0.23)
SNA Change, v	0.36***	0.36***	0.50***	1.33***	1.19***	1.67***
	(0.03)	(0.03)	(0.05)	(0.11)	(0.12)	(0.17)
Data Management Error, v	6.29***	6.18***		12.95***	11.09***	14.95***
	(1.14)	(1.12)		(1.61)	(1.53)	(1.90)
Log Difference, $v-y$	−1.12***	−1.01***	−1.32***	−4.78***	−4.58***	−4.01***
	(0.03)	(0.04)	(0.05)	(0.11)	(0.12)	(0.18)
Intercept	−2.22***	−2.65***	−2.16***	1.08	−0.61	1.80**
	(0.56)	(0.56)	(0.62)	(0.85)	(0.90)	(0.91)
AIC	137257.77	116430.55	51510.37	390633.64	333012.01	154288.32
Log Likelihood	−68615.88	−58202.28	−25742.18	−195302.82	−166492.01	−77130.16
Observations	349493	295467	135270	58567	49968	23226
Number of Countries	162	135	66	162	135	66
Number of Reference Years	27	31	24	27	31	24
Number of Vintages	103	103	103	76	78	71
Variance: Country (Intercept)	0.24	0.24	0.30	3.20	3.42	4.91
Variance: Reference Year (Intercept)	0.08	0.03	0.16	7.51	11.78	2.34
Variance: Vintage (Intercept)	27.48	27.25	27.67	26.63	22.30	15.83

Notes: This table presents three logistic regressions (Models 1–3) and three linear regressions (Models 4–6) with random intercepts for country, year, and vintage. Models 1 and 4 replace *Public Administration* with the Hanson and Sigman (2021) *State Capacity* index; Models 2 and 5 with the ICRG measure of *Bureaucratic Quality*; Models 3 and 6 with the Brambor et al. (2020) *Information Capacity* index, which covers 85 countries and up to 2012 only. After merging with the other covariates, Models 3 and 6 are estimated on 66 countries. *Data Management Error* is excluded from Model 3 due to near-perfect separation in the restricted sample. y is the reference year; v is the vintage. Standard errors in parentheses. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

magnitude regardless of which capacity measure is used, and the coefficients are broadly consistent with the main results. The capacity measures themselves show mixed results: *Bureaucratic Quality* is significant for both outcomes, *State Capacity* is insignificant throughout, and *Information Capacity* is insignificant for likelihood but negative and significant for magnitude. This last result should be interpreted with caution given the substantially restricted sample (66 countries, up to 2012). *Data Management Error* is excluded from Model 3 due to near-perfect separation in this restricted sample.

D.2 Economic Development and EU Membership

In Table D.2, Models 1 and 3 add World Bank income group as a control for economic development, which is bound to affect statistical capacity. This reference-year variable, which is lagged, ranges from 1 to 4, covering the World Bank's four income groups: low income, lower middle income, upper middle income, and high income. The democracy finding is robust: *Polyarchy* remains positive and significant for both outcomes (0.55 and 1.17, both p¡0.01), essentially unchanged from the main models. The analysis is also robust to the possibility that EU membership generates distinct supranational scrutiny. Controlling for EU membership in Models 2 and 4 of Table D.2 leaves the democracy finding essentially unchanged. Importantly, *EU Membership* takes the value of 1 only for member-years and 0 otherwise (before accession or after leaving).

Table D.2: Determinants of the Likelihood and Magnitude of Data Revisions: Controlling for World Bank Income Group and European Union Membership

	Dependent Variable:			
	Revision	Revision	Abs. % Change	Abs. % Change
	(1)	(2)	(3)	(4)
Polyarchy, $y-1$	0.55***	0.57***	1.17***	1.28***
	(0.09)	(0.09)	(0.34)	(0.34)
Income Group, $y-1$	0.03*		0.46***	
	(0.02)		(0.07)	
EU Membership, $y-1$		−0.15***		0.94***
		(0.04)		(0.18)
Public Administration, $y-1$	0.03	0.04**	0.07	0.10
	(0.02)	(0.02)	(0.08)	(0.08)
IMF Program, $y-1$	0.02	0.01	−0.07	−0.10
	(0.02)	(0.02)	(0.08)	(0.08)
Financial Crisis, $y-1$	0.00	0.00	−0.10	−0.17**
	(0.02)	(0.02)	(0.08)	(0.08)
Natural Disaster, $y-1$	0.06***	0.05***	−0.10	−0.11
	(0.02)	(0.02)	(0.07)	(0.07)
Armed Conflict, $y-1$	0.05	0.05*	−0.21	−0.25*
	(0.03)	(0.03)	(0.13)	(0.13)
SNA Change, v	0.36***	0.36***	1.25***	1.26***
	(0.03)	(0.03)	(0.11)	(0.11)
Data Management Error, v	6.27***	6.29***	11.57***	11.55***
	(1.15)	(1.16)	(1.49)	(1.49)
Log Difference, $v-y$	−1.02***	−1.02***	−4.67***	−4.66***
	(0.03)	(0.03)	(0.10)	(0.10)
Intercept	−2.53***	−2.48***	−1.20	−0.33
	(0.56)	(0.56)	(0.94)	(0.92)
AIC	148060.59	148366.52	419515.95	420273.84
Log Likelihood	−74016.30	−74169.26	−209742.97	−210121.92
Observations	374619	375247	62934	63050
Number of Countries	169	169	169	169
Number of Reference Years	31	31	31	31
Number of Vintages	103	103	78	78
Variance: Country (Intercept)	0.23	0.24	2.55	2.76
Variance: Reference Year (Intercept)	0.05	0.04	13.29	12.88
Variance: Vintage (Intercept)	28.13	27.91	26.14	26.20

Notes: This table presents two logistic regressions (Models 1–2) and two linear regressions (Models 3–4) with random intercepts for country, year, and vintage. Models 1 and 3 include *Income Group*. Models 2 and 4 include *EU Member*. y is the reference year; v is the vintage. Standard errors in parentheses. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

D.3 Excluding EU Member-Years

The main findings also hold when EU member-years are excluded from the sample in Table D.3. In this specification, I exclude observations during EU membership rather than excluding EU countries entirely: accession countries contribute their pre-membership years, and only member-years are dropped. EU membership is associated with less frequent but larger revisions, consistent with standardized Eurostat reporting that reduces routine revisions while concentrating methodological ones. Still, this does not account for the democracy result, which remains robust.

Table D.3: Determinants of the Likelihood and Magnitude of Data Revisions: Excluding European Union Member-Years

	Dependent Variable:	
	Revision	Abs. % Change
	(1)	(2)
Polyarchy, $y-1$	0.44^{***}	1.02^{***}
	(0.09)	(0.35)
Public Administration, $y-1$	0.03^{*}	0.11
	(0.02)	(0.08)
IMF Program, $y-1$	0.04^{**}	-0.11
	(0.02)	(0.08)
Financial Crisis, $y-1$	0.04^{**}	-0.18^{**}
	(0.02)	(0.09)
Natural Disaster, $y-1$	0.05^{**}	-0.13^{*}
	(0.02)	(0.08)
Armed Conflict, $y-1$	0.05	-0.27^{*}
	(0.03)	(0.14)
SNA Change, v	0.29^{***}	1.03^{***}
	(0.03)	(0.12)
Data Management Error, v	6.25^{***}	11.69^{***}
	(1.13)	(1.51)
Log Difference, $v-y$	-1.01^{***}	-5.00^{***}
	(0.04)	(0.11)
Intercept	-2.37^{***}	0.42
	(0.55)	(0.98)
AIC	130227.48	366335.16
Log Likelihood	−65100.74	−183153.58
Observations	331668	54668
Number of Countries	157	157
Number of Reference Years	31	31
Number of Vintages	103	78
Variance: Country (Intercept)	0.24	2.73
Variance: Reference Year (Intercept)	0.03	14.98
Variance: Vintage (Intercept)	26.71	28.97

Notes: This table presents one logistic regression (Model 1) and one linear regression (Model 3) with random intercepts for country, year, and vintage. y is the reference year; v is the vintage. Standard errors in parentheses. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

D.4 World Bank and IMF Lending

Table D.4 addresses a possibility raised by Hollyer et al. (2025): pressure from the Bretton Woods institutions might drive data revision. I replace *IMF Program* (used in the main models) with the volume of World

Bank and IMF lending, measured as the logged value of commitments (Models 1 and 3) and disbursements (Models 2 and 4). Hollyer et al.'s (2025) analysis covered 1980–2010; following the authors' procedures, I re-constructed these variables from the raw data for my analysis period, 1990–2020. The democracy finding is unchanged: *Polyarchy* remains positive and significant for both the likelihood and magnitude of revision across all specifications. World Bank lending is associated with marginally smaller revisions in magnitude, but IMF lending is insignificant throughout, and neither institution's involvement accounts for the democracy result.

Table D.4: Determinants of the Likelihood and Magnitude of Data Revisions: Controlling for World Bank and IMF Lending

	Dependent Variable:			
	Revision	Revision	Abs. % Change	Abs. % Change
	(1)	(2)	(3)	(4)
Polyarchy, $y-1$	0.57***	0.57***	1.53***	1.55***
	(0.09)	(0.09)	(0.34)	(0.34)
Log WB Commitment, $y-1$	0.00		−0.04***	
	(0.00)		(0.00)	
Log WB Disbursement, $y-1$		0.00		−0.03***
		(0.00)		(0.00)
Log IMF Commitment, $y-1$	−0.00		0.00	
	(0.00)		(0.00)	
Log IMF Disbursement, $y-1$		−0.00		−0.01
		(0.00)		(0.00)
Public Administration, $y-1$	0.04*	0.04*	0.10	0.11
	(0.02)	(0.02)	(0.08)	(0.08)
Financial Crisis, $y-1$	0.00	0.01	−0.17**	−0.16**
	(0.02)	(0.02)	(0.08)	(0.08)
Natural Disaster, $y-1$	0.05***	0.05***	−0.10	−0.10
	(0.02)	(0.02)	(0.07)	(0.07)
Armed Conflict, $y-1$	0.05*	0.05*	−0.27**	−0.26**
	(0.03)	(0.03)	(0.13)	(0.13)
SNA Change, v	0.36***	0.36***	1.26***	1.26***
	(0.03)	(0.03)	(0.11)	(0.11)
Data Management Error, v	6.29***	6.29***	11.46***	11.45***
	(1.16)	(1.16)	(1.49)	(1.49)
Log Difference, $v-y$	−1.02***	−1.02***	−4.66***	−4.66***
	(0.03)	(0.03)	(0.10)	(0.10)
Intercept	−2.48***	−2.49***	−0.03	−0.05
	(0.56)	(0.56)	(0.91)	(0.91)
AIC	148379.24	148378.81	420242.77	420243.19
Log Likelihood	−74175.62	−74175.40	−210106.38	−210106.59
Observations	375247	375247	63050	63050
Number of Countries	169	169	169	169
Number of Reference Years	31	31	31	31
Number of Vintages	103	103	78	78
Variance: Country (Intercept)	0.23	0.23	2.76	2.74
Variance: Reference Year (Intercept)	0.05	0.05	12.57	12.52
Variance: Vintage (Intercept)	27.89	27.89	26.20	26.18

Notes: This table presents two logistic regressions (Models 1–2) and two linear regressions (Models 3–4) with random intercepts for country, year, and vintage. Models 1 and 3 control for World Bank and IMF commitments. Models 2 and 4 control for World Bank and IMF disbursements. y is the reference year; v is the vintage. Standard errors in parentheses. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

D.5 Dichotomized Vintage Regime

In the main text, I examine the timing of democracy using the continuous *Polyarchy* measure at both the reference year (y) and the vintage year (v). Table D.5 reports robustness checks that instead classify the vintage regime as democratic or autocratic. To do so, I use the V-Dem Regimes of the World typology (Lührmann et al., 2018), which sorts country-years into closed autocracies, electoral autocracies, electoral democracies, and liberal democracies; I code the latter two as democratic. This typology draws on the same underlying V-Dem data as Polyarchy, but combines several indicators of electoral competition and freedom to classify regimes categorically, rather than relying on a single threshold.

Table D.5: Determinants of the Likelihood and Magnitude of Data Revisions: Controlling for Vintage-Year Democracy (Dichotomous Measure)

	Dependent Variable:					
	Revision	Revision	Revision	Abs. % Change	Abs. % Change	Abs. % Change
	(1)	(2)	(3)	(4)	(5)	(6)
Polyarchy, $y-1$	0.59***	0.33**	0.69***	1.13**	0.27	−0.46
	(0.12)	(0.15)	(0.12)	(0.47)	(0.59)	(0.48)
Democracy, $v-1$			0.01			−0.61**
			(0.07)			(0.27)
Polyarchy, $y-1 \times$ Democracy, $v-1$			−0.15			2.44***
			(0.13)			(0.54)
Public Administration, $y-1$	0.05*	0.04	0.04**	0.09	0.19*	0.09
	(0.03)	(0.03)	(0.02)	(0.12)	(0.10)	(0.08)
IMF Program, $y-1$	0.01	0.05*	0.02	−0.01	−0.18	−0.12
	(0.03)	(0.03)	(0.02)	(0.10)	(0.11)	(0.08)
Financial Crisis, $y-1$	−0.03	0.09***	−0.00	0.10	−0.43***	−0.16**
	(0.03)	(0.03)	(0.02)	(0.10)	(0.13)	(0.08)
Natural Disaster, $y-1$	0.05**	0.03	0.05***	−0.04	−0.28**	−0.11
	(0.02)	(0.03)	(0.02)	(0.09)	(0.11)	(0.07)
Armed Conflict, $y-1$	−0.04	0.12**	0.05	−0.62***	−0.03	−0.18
	(0.05)	(0.05)	(0.03)	(0.19)	(0.19)	(0.13)
SNA Change, v	0.40***	0.33***	0.36***	1.45***	0.78***	1.24***
	(0.04)	(0.04)	(0.03)	(0.15)	(0.17)	(0.11)
Data Management Error, v			6.29***			11.47***
			(1.16)			(1.49)
Log Difference, $v-y$	−1.04***	−0.93***	−1.02***	−4.03***	−5.20***	−4.65***
	(0.04)	(0.02)	(0.03)	(0.13)	(0.17)	(0.10)
Intercept	−2.61***	−2.74***	−2.49***	−1.10	0.30	0.08
	(0.59)	(0.61)	(0.56)	(0.88)	(1.11)	(0.92)
AIC	84194.81	63022.10	148375.47	239915.49	179168.45	420269.72
Log Likelihood	−42085.40	−31499.05	−74172.74	−119944.74	−89571.22	−210118.86
Observations	204147	171100	375247	36313	26737	63050
Number of Countries	112	110	169	112	110	169
Number of Reference Years	31	31	31	31	31	31
Number of Vintages	103	103	103	73	73	78
Variance: Country (Intercept)	0.28	0.39	0.23	4.50	3.80	2.88
Variance: Reference Year (Intercept)	0.08	0.00	0.04	8.65	15.98	12.64
Variance: Vintage (Intercept)	28.78	30.64	27.90	21.07	39.23	26.30

Notes: This table presents three logistic regressions (Models 1–3) and three linear regressions (Models 4–6) with random intercepts for country, year, and vintage. All models include the dichotomous variable *Democracy*, referring to regime type at $v-1$; Models 3 and 6 additionally include an interaction between *Polyarchy* and *Democracy*. *Data Management Error* is excluded from Models 1–2 and 4–5 due to near-perfect separation in the restricted samples. y is the reference year; v is the vintage. Standard errors in parentheses. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

The first check (Models 1–2 and 4–5) estimates the association between reference-year democracy and

revision separately within autocratic-vintage and democratic-vintage country-years. The second (Models 3 and 6) interacts continuous reference-year democracy with this binary vintage classification. Both corroborate the continuous interaction reported in the main text: reference-year democracy predicts the likelihood of revision regardless of the vintage regime, while its association with revision magnitude is concentrated among democratic-vintage observations. Note that *Data Management Error* is omitted from the split-sample models, where it perfectly predicts the outcome within the democratic vintage subsample; it is retained in the interaction models, which are estimated on the full sample.

E Reduced Sample: Only Main Releases

The WDI are typically updated twice a year, with additional updates as needed. As the World Bank shifted from print to digital, the number of annual updates increased: there were ten updates in 2017 (in every month except for January and February), compared to only one each year from 1997 to 2004 (always in April). While the main models include data on all available vintages, Table E.1 uses only each year's main scheduled release, typically in April.[2] Restricting the analysis to the main release reduces noise (since the information provided by two consecutive releases is likely very similar) and ensures that recent years (with many releases) do not disproportionately influence the results compared to earlier years (with fewer releases).

The democracy finding holds for likelihood across both specifications, but the magnitude results are weaker in this restricted sample, likely reflecting the substantial reduction in observations and vintage variation. *Data Management Error* is excluded from the likelihood models due to near-perfect separation in the restricted sample, but retained in the magnitude models where it is positive but insignificant, consistent with the small number of documented coding errors among the main annual releases.

[2]The analysis accommodates two exceptions: due to pandemic-related delays in data collection and dissemination, the main release for 2020 was in August, and the main release for 2021 was in September.

Table E.1: Determinants of the Likelihood and Magnitude of Data Revisions: Only Main Releases

	Dependent Variable:			
	Revision	Revision	Abs. % Change	Abs. % Change
	(1)	(2)	(3)	(4)
Polyarchy, $y-1$	0.67^{***}		0.40	
	(0.18)		(0.50)	
Media Freedom, $y-1$		0.06^{*}		0.27^{***}
		(0.03)		(0.10)
Freedom of Academic Exchange, $y-1$		0.16^{***}		-0.43^{***}
		(0.05)		(0.15)
Suffrage, $y-1$		0.10		1.07
		(0.26)		(0.71)
Opposition Strength, $y-1$		-0.29^{**}		-1.44^{***}
		(0.12)		(0.35)
Turnover Frequency, Prev. 5 Yrs, $y-1$		0.02		-0.14
		(0.03)		(0.09)
Political Corruption Index, $y-1$		-0.28		-1.94^{***}
		(0.21)		(0.57)
Public Administration, $y-1$	0.08^{**}	0.04	0.35^{***}	0.29^{**}
	(0.04)	(0.05)	(0.12)	(0.14)
IMF Program, $y-1$	0.04	0.05	-0.12	-0.10
	(0.04)	(0.04)	(0.13)	(0.13)
Financial Crisis, $y-1$	0.11^{**}	0.08^{*}	-0.23^{*}	-0.18
	(0.04)	(0.05)	(0.13)	(0.14)
Natural Disaster, $y-1$	0.01	0.04	-0.11	-0.13
	(0.04)	(0.04)	(0.11)	(0.12)
Armed Conflict, $y-1$	0.02	0.03	0.05	0.14
	(0.07)	(0.08)	(0.21)	(0.23)
SNA Change, v	-0.22^{***}	-0.18^{**}	1.48^{***}	1.56^{***}
	(0.08)	(0.08)	(0.18)	(0.19)
Data Management Error, v			13.91^{**}	12.77^{**}
			(6.18)	(6.17)
Log Difference, $v-y$	-1.35^{***}	-1.26^{***}	-4.56^{***}	-4.50^{***}
	(0.07)	(0.08)	(0.20)	(0.21)
Intercept	1.22	1.06	-0.29	0.78
	(0.88)	(0.96)	(1.10)	(1.33)
AIC	31247.48	28664.25	126109.23	114327.36
Log Likelihood	-15611.74	-14315.13	-63040.62	-57144.68
Observations	66973	61611	19397	17586
Number of Countries	169	166	168	164
Number of Reference Years	30	29	28	27
Number of Vintages	25	25	23	23
Variance: Country (Intercept)	0.60	0.64	3.10	3.26
Variance: Reference Year (Intercept)	0.58	0.42	8.54	8.19
Variance: Vintage (Intercept)	17.81	19.46	17.18	16.64

Notes: This table presents two logistic regressions (Models 1–2) and two linear regressions (Models 3–4) with random intercepts for country, year, and vintage, estimated on the subset of vintages corresponding to the main scheduled annual release (typically April; August for 2020 and September for 2021 due to pandemic-related delays). *Data Management Error* is excluded from Models 1–2 due to near-perfect separation in the restricted sample but is retained in the magnitude models (Models 3–4). y is the reference year; v is the vintage. Standard errors in parentheses. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

F Models With Fixed Effects

F.1 Fixed Effects for Country, Reference Year, and Vintage

Table F.1: Determinants of the Likelihood and Magnitude of Data Revisions: Fixed Effects for Country, Reference Year, and Vintage

	Dependent Variable:			
	Revision	Revision	Abs. % Change	Abs. % Change
	(1)	(2)	(3)	(4)
Polyarchy, $y-1$	0.52**		0.69	
	(0.24)		(0.72)	
Media Freedom, $y-1$		0.05*		0.33***
		(0.03)		(0.12)
Freedom of Academic Exchange, $y-1$		0.09		−0.26
		(0.06)		(0.24)
Suffrage, $y-1$		0.09		0.78
		(0.36)		(1.04)
Opposition Strength, $y-1$		−0.04		−0.78**
		(0.10)		(0.37)
Turnover Frequency, Prev. 5 Yrs, $y-1$		0.00		0.07
		(0.02)		(0.10)
Political Corruption Index, $y-1$		0.37		0.16
		(0.25)		(0.93)
Public Administration, $y-1$	0.02	0.02	−0.01	−0.01
	(0.05)	(0.05)	(0.13)	(0.13)
IMF Program, $y-1$	0.02	0.02	−0.04	−0.02
	(0.03)	(0.03)	(0.13)	(0.14)
Financial Crisis, $y-1$	−0.01	−0.02	−0.13	−0.16
	(0.04)	(0.04)	(0.14)	(0.14)
Natural Disaster, $y-1$	0.05**	0.06***	−0.13*	−0.14*
	(0.02)	(0.02)	(0.07)	(0.07)
Armed Conflict, $y-1$	0.02	0.01	−0.31	−0.22
	(0.07)	(0.07)	(0.27)	(0.31)
SNA Change, v	0.35***	0.37***	1.21***	1.25***
	(0.10)	(0.10)	(0.37)	(0.37)
Data Management Error, v	6.41***	13.59***	11.75***	11.33***
	(0.58)	(0.16)	(1.00)	(1.00)
Observations	267850	248886	63048	58718
Number of Countries	169	166	169	165
Number of Reference Years	31	30	31	30
Number of Vintages	78	77	76	76
Deviance	148645.03	139091.68		
Log Likelihood	−74322.52	−69545.84		
R^2 (Within)			0.00	0.00

Notes: This table presents two logistic regressions (Models 1–2) and two linear regressions (Models 3–4) with fixed effects for country, year, and vintage. Standard errors clustered by country. The likelihood models drop observations from groups with no within-group variation in the outcome (incidental parameters problem), reducing the sample. *Log Difference* is excluded from all models due to collinearity with vintage fixed effects. y is the reference year; v is the vintage. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

Table F.1 replaces the random effects specification with country, year, and vintage fixed effects. Given the large number of units (169 countries), time periods (31 years), and vintages (103), these models are computationally inefficient: they lead to a substantial loss of degrees of freedom, and *Log Difference* is collinear with the vintage fixed effects. The models are not directly comparable to the random effects models because of the reduced sample size. This problem is especially pronounced for the likelihood models

(Models 1 and 2), which suffer from the incidental parameters problem: observations from groups with no within-group variation in the outcome are automatically dropped. Fixed effects models also discard between-country variation and focus exclusively on within-country changes over time, which is particularly costly for slowly-changing variables like democracy.

Despite these limitations, the main democracy finding survives for likelihood: *Polyarchy* is positive and significant in Model 1 ($p < 0.05$). The magnitude result is less robust, which is expected given that the between-country variation in democracy is absorbed by the country fixed effects. Among the mechanism indicators, *Media Freedom* is significant for magnitude ($p < 0.01$) but not for likelihood, while *Freedom of Academic Exchange* is never significant.

F.2 Fixed Effects for Reference Year Only

Table F.2 reports a specification with year fixed effects only, omitting the country and vintage effects. This directly addresses whether common time shocks (such as changes to internationally advised best practices for GDP calculation) drive the results. Because this specification retains between-country variation (which the country fixed effects in Table F.1 absorb), it recovers the cross-sectional association between democracy and revision that is central to my argument. The democracy finding is robust: *Polyarchy* is positive and significant for both the likelihood ($p < 0.01$) and magnitude ($p < 0.01$) of revision. As in the main analysis, the mechanism indicators are more mixed: *Media Freedom* is significant for likelihood but not magnitude, while *Freedom of Academic Exchange* is never significant.

Table F.2: Determinants of the Likelihood and Magnitude of Data Revisions: Fixed Effects for Reference Year Only

	Dependent Variable:			
	Revision (1)	Revision (2)	Abs. % Change (3)	Abs. % Change (4)
Polyarchy, $y-1$	0.36***		3.21***	
	(0.09)		(0.68)	
Media Freedom, $y-1$		0.05**		0.23
		(0.02)		(0.16)
Freedom of Academic Exchange, $y-1$		0.01		−0.03
		(0.03)		(0.20)
Suffrage, $y-1$		0.13		1.75***
		(0.10)		(0.63)
Opposition Strength, $y-1$		−0.03		1.51***
		(0.06)		(0.56)
Turnover Frequency, Prev. 5 Yrs, $y-1$		0.00		−0.11
		(0.01)		(0.15)
Political Corruption Index, $y-1$		−0.31***		−3.57***
		(0.10)		(0.76)
Public Administration, $y-1$	0.05**	0.01	0.68***	0.21
	(0.02)	(0.03)	(0.18)	(0.21)
IMF Program, $y-1$	0.00	0.02	−0.94***	−0.60**
	(0.03)	(0.02)	(0.23)	(0.23)
Financial Crisis, $y-1$	0.01	0.01	−0.34	−0.17
	(0.03)	(0.03)	(0.22)	(0.21)
Natural Disaster, $y-1$	0.02	0.04*	0.08	0.23
	(0.02)	(0.02)	(0.18)	(0.17)
Armed Conflict, $y-1$	0.12***	0.12***	0.42	0.43
	(0.04)	(0.04)	(0.32)	(0.30)
SNA Change, v	0.37***	0.37***	0.61	0.63
	(0.07)	(0.07)	(0.45)	(0.46)
Data Management Error, v	4.94***	14.01***	18.60***	18.16***
	(1.28)	(0.06)	(1.02)	(1.10)
Log Difference, $v-y$	−1.51***	−1.48***	−4.58***	−4.52***
	(0.02)	(0.02)	(0.16)	(0.17)
Observations	375247	350529	63050	58721
Number of Reference Years	31	30	31	30
Deviance	305719.71	285601.18		
Log Likelihood	−152859.85	−142800.59		
R^2 (Within)			0.14	0.14

Notes: This table presents two logistic regressions (Models 1–2) and two linear regressions (Models 3–4) with fixed effects for year. Standard errors clustered by The likelihood models drop observations from groups with no within-group variation in the outcome (incidental parameters problem), reducing the sample. y is the reference year; v is the vintage. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

G Direction of Revisions

Table G.1 examines the direction of revisions, asking whether democracies are more likely to revise their GDP estimates upward rather than downward. The analysis is restricted to observations where a revision occurred and neither GDP value is missing, excluding appearances and disappearances where direction is undefined. Model 1 shows that higher Polyarchy scores are associated with a significantly higher probability of upward revisions ($p < 0.05$), consistent with the expectation that democratic governments initially publish cautious, understated estimates that are subsequently corrected upward as better data becomes available. This result is also consistent with Martínez's (2022) finding that autocrats tend to inflate their numbers

Table G.1: Direction of Revisions

	Dependent Variable:	
	Upward Revision	Upward Revision
	(1)	(2)
Polyarchy, $y-1$	0.16^{**}	
	(0.07)	
Media Freedom, $y-1$		0.01
		(0.02)
Freedom of Academic Exchange, $y-1$		0.01
		(0.02)
Suffrage, $y-1$		-0.09
		(0.09)
Opposition Strength, $y-1$		0.01
		(0.06)
Turnover Frequency, Prev. 5 Yrs, $y-1$		-0.01
		(0.02)
Political Corruption Index, $y-1$		-0.13^{*}
		(0.08)
Public Administration, $y-1$	0.01	0.00
	(0.02)	(0.02)
IMF Program, $y-1$	0.00	0.01
	(0.02)	(0.02)
Financial Crisis, $y-1$	-0.05^{**}	-0.03
	(0.02)	(0.02)
Natural Disaster, $y-1$	0.00	0.01
	(0.02)	(0.02)
Armed Conflict, $y-1$	-0.01	-0.01
	(0.03)	(0.03)
SNA Change, v	0.19^{***}	0.20^{***}
	(0.03)	(0.03)
Data Management Error, v	0.78	0.94^{*}
	(0.49)	(0.52)
Log Difference, $v-y$	-0.03^{***}	-0.04^{***}
	(0.01)	(0.01)
Intercept	0.23^{***}	0.43^{***}
	(0.07)	(0.13)
AIC	84278.44	78452.20
Log Likelihood	-42126.22	-39208.10
Observations	63050	58721
Number of Countries	169	166
Number of Reference Years	31	30
Number of Vintages	78	78
Variance: Country (Intercept)	0.02	0.02
Variance: Reference Year (Intercept)	0.00	0.00
Variance: Vintage (Intercept)	0.23	0.22

Notes: Logistic regressions with random intercepts for country, year, and vintage, estimated on observations where a revision occurred and neither GDP value is missing (*Abs. % Change* > 0). The outcome equals 1 if the revision is upward and 0 if downward. Appearances and disappearances are excluded since the direction of revision is undefined when one value is missing. y is the reference year; v is the vintage. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.1$.

from the outset.

Model 2 decomposes this finding across the democracy mechanism indicators, but none reach conventional significance levels, suggesting that the upward revision tendency is a property of democratic governance broadly rather than any specific institutional channel. SNA changes are consistently associated with upward revisions across both models, reflecting that methodological updates to national accounts (such as the adoption of new SNA standards or the inclusion of previously unmeasured economic activities) tend to increase

GDP estimates rather than reduce them. Relatedly, the negative and significant coefficient on *Log Difference* suggests that upward revisions are disproportionately concentrated in the period shortly after initial data release, when new survey information and administrative data are most likely to prompt upward corrections to initially cautious estimates. As more time passes, the remaining revisions become less systematically directional.

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